



3800 Finerty Road  
Victoria, BC

April 2, 2026

**Facilities Management**

The University of Victoria  
3800 Finerty Road  
Victoria, BC

Dear Facilities Management,

We are the BuildHers: Ayla, Claire, Julia, and Sabrina. Please find our enclosed report titled “Recommended Solution for Universally Accessible Smart Building Management at UVic,” prepared in response to RFP-202601.

The attached report addresses the lack of centralized control across the University of Victoria’s building systems. The report presents the design and testing of three prototype concepts: SmartSafe, SmartClimate, and SmartZone. SmartSafe links fire detection and accessible emergency lighting, SmartClimate optimizes HVAC performance, and SmartZone provides occupancy-based control of HVAC and lighting systems. We evaluated each design using a Raspberry Pi Pico W and a building zone representative breadboard. Based on testing results and secondary research, we recommend an integrated solution that combines the strongest features of each design while prioritizing accessibility.

Thank you for the opportunity to develop and present this recommendation. We welcome your feedback and look forward to hearing from you regarding the next steps for development.

Sincerely,

Ayla Sorensen

Claire Saunders-Kong

Julia Kalhous

Sabrina Bryce

**The BuildHers**

Attachment: Recommended Solution for Universally Accessible Smart Building Management at UVic



# Recommended Solution for Universally Accessible Smart Building Management at UVic

**In Response to RFP-202601**

**Submitted to  
UVic Facilities Management**

**Submitted by The BuildHers:  
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**Date  
April 2<sup>nd</sup>, 2026**

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## Executive Summary

Facilities Management at the University of Victoria is seeking a solution to the lack of integrated and accessible building systems management. HVAC, lighting, security, and fire safety systems currently operate independently of each other. This fragmented management limits centralized control, reducing operational efficiency while restricting accessibility for users.

To address this, we designed three systems and developed prototypes using a Raspberry Pi Pico W and a building zone representative breadboard: SmartSafe, SmartClimate, and SmartZone. SmartSafe provides multi-sensor fire detection and emergency lighting to improve reliability and exit visibility in emergency conditions. SmartClimate monitors temperature and occupancy to optimize HVAC performance and improve occupant comfort in UVic's older buildings. SmartZone uses occupancy-based control of HVAC and lighting to reduce energy consumption through dynamic environmental response.

We tested each prototype through repeated laboratory trials and supported our findings with secondary research. To enable direct comparison, we defined quantitative performance metrics and applied standardized scoring criteria. We prioritized accessibility throughout the evaluation and assessed each design's user interface against the 7 Principles of Universal Design. Primary testing measured system response accuracy to environmental inputs, the consistency of user interface updates, and the number of functional systems integrated. We evaluated scalability and system longevity using per-zone sensor cost calculations and MVP component counts, and we estimated projected energy savings using supporting research. The results show that SmartZone provides the strongest foundation for integration and energy efficiency, while SmartSafe strengthens system reliability in emergency scenarios, and SmartClimate offers a cost-effective approach to thermal control. However, no single design satisfies all objectives. We therefore recommend the further development of an integrated system that combines the strengths of all three designs to provide an optimized solution.

## Glossary

|                                                   |                                                                                    |
|---------------------------------------------------|------------------------------------------------------------------------------------|
| User Interface (UI)                               | Communication point between a device and its user.                                 |
| Minimum Viable Product (MVP)                      | Simplest version of a product that teams use to test and validate a design.        |
| Building Management System (BMS)                  | Computer-based control framework for building systems.                             |
| Light emitting diode (LED)                        | Semiconductor device that emits light when an electrical current runs through it.  |
| CdS (Cadmium Sulfide) Photoresistor               | Light-dependent resistor that decreases resistance when light levels increase.     |
| Negative Temperature Coefficient (NTC) Thermistor | Semiconductor resistor that decreases in resistance when temperature increases.    |
| Infrared (IR) Light                               | Non-visible light that is part of the electromagnetic spectrum.                    |
| Phototransistor                                   | Semiconductor device that turns light into electricity.                            |
| Potentiometer                                     | Manually adjustable tri-terminal resistor that acts as a variable voltage divider. |
| Domestic Hot Water Generator (DHW)                | Water heating system commonly implemented for domestic use.                        |
| Passive Infrared (PIR) Motion Sensor              | Detects infrared light emitted by warm bodies.                                     |

## 1. Fragmented Building Management at UVic

The University of Victoria (UVic) Facilities Management oversees the operation of campus building systems: heating, ventilation, and air conditioning (HVAC), lighting, fire alarm, and security. Currently, these systems function independently with separate hardware and user interfaces (UIs), limiting centralized control. In response, UVic Facilities Management released RFP-202601 [1], calling for the development of a minimum viable product (MVP) that uses a Raspberry Pi Pico W to demonstrate the control of a building system and the capacity for future integration across all building systems.

### 1.1 Need for Integrated Building Control

Current building management systems at UVic provide insufficient unified control; Facilities Management needs the ability to monitor and regulate building systems centrally.

### 1.2 Current Goal

The goal is centralized building management and intelligent system operation at UVic.

### 1.3 Performance Objectives

**Table 1** outlines the performance objectives that govern design evaluation.

**Table 1. Design Objectives and Measurement Methods**

| Objective                           | Basis for Measurement                             | Units               |
|-------------------------------------|---------------------------------------------------|---------------------|
| Ensure accessibility                | Universal Design principles satisfied             | # Principles        |
| Enable system integration           | Distinct building systems controlled              | # Systems           |
| Support scalability                 | Per-zone system sensors cost                      | \$ CAD / Zone       |
| Provide real-time visibility        | Correct UI updates within programmed timeframe    | % Timely Updates    |
| Respond to environmental conditions | Correct system response within defined parameters | % Correct Responses |
| Improve system efficiency           | Potential reduction in system energy use          | % Savings           |
| Maintain system longevity           | Number of MVP components used                     | # Components        |

### 1.4 Constraints

In accordance with RFP-202601 and laboratory guidelines [2], the designs must

- Use a Raspberry Pi Pico W as the only microcontroller
- Use MicroPython as the programming language
- Incorporate a minimum of three sensors (with at least two different types)
- Be feasible within laboratory timelines and hardware constraints
- Include a user-friendly UI that allows for wireless system interaction
- Represent building-level rather than single room system control
- Be complete by April 2, 2026

## 1.5 The BuildHers Approach to UVic Building Systems Integration

The BuildHers aim to provide UVic with a building management system (BMS) design that is accessible to all users. The team ultimately envisions integrated control of all campus building operations through a universally accessible UI. **Figure 1** illustrates the projected full-scale solution.

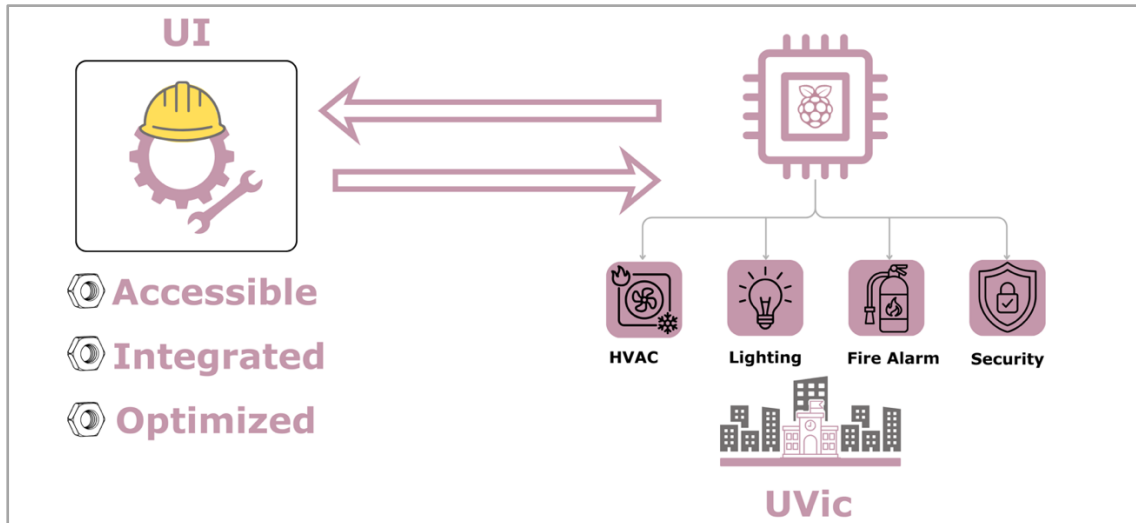


Figure 1. Illustration of Ultimate BMS Solution

An MVP design attests to the solution’s viability. The successful activation of light-emitting diodes (LEDs) demonstrates the response of actuator systems, while the preliminary UI illustrates user-friendly central control of a single building. This verifies the feasibility of providing accessible monitoring across campus when scaled.

The following sections outline three proposed solutions, present a comparative analysis, and provide a final recommendation for integrated and inclusive building management at UVic.

## 2. Proposed Design Concepts and Development Plans

The BuildHers built and tested three MVP designs to evaluate their potential for accessible centralized building management:

- **SmartSafe:** An integrated fire detection and emergency lighting system built to improve accessibility for building users during emergencies.
- **SmartClimate:** An HVAC optimization system that aims to improve thermal comfort for occupants in aging campus buildings.
- **SmartZone:** An integrated occupancy-based HVAC and lighting system that acts to improve the energy efficiency of campus operations.

In each prototype, a Raspberry Pi Pico W serves as the central controller, processing sensor inputs and hosting a wireless web-based UI. A representative breadboard circuit shows the viability of building control. All designs use materials within the project kit (initial cost: \$30). **Table 2** outlines its contents.

**Table 2. Common Design Hardware**

| Component                | Quantity | Data Sheet Reference |
|--------------------------|----------|----------------------|
| 830 Tie-Point Breadboard | 1        | [3]                  |
| Raspberry Pi Pico W      | 1        | [4]                  |
| CdS Photoresistor        | 1        | [5]                  |
| NTC Thermistor           | 1        | [6]                  |
| Infrared (IR) LED        | 2        | [7]                  |
| IR Phototransistor       | 2        | [8]                  |
| Red LED                  | 1        | [9]                  |
| Green LED                | 1        | [10]                 |
| Potentiometer            | 1        | [11]                 |

Each prototype underwent final laboratory testing. To provide a common means of comparison, the team established scoring tables that a) evaluate design accessibility, b) present primary research, and c) provide secondary research. For the detailed testing procedure, refer to **Appendix B**.

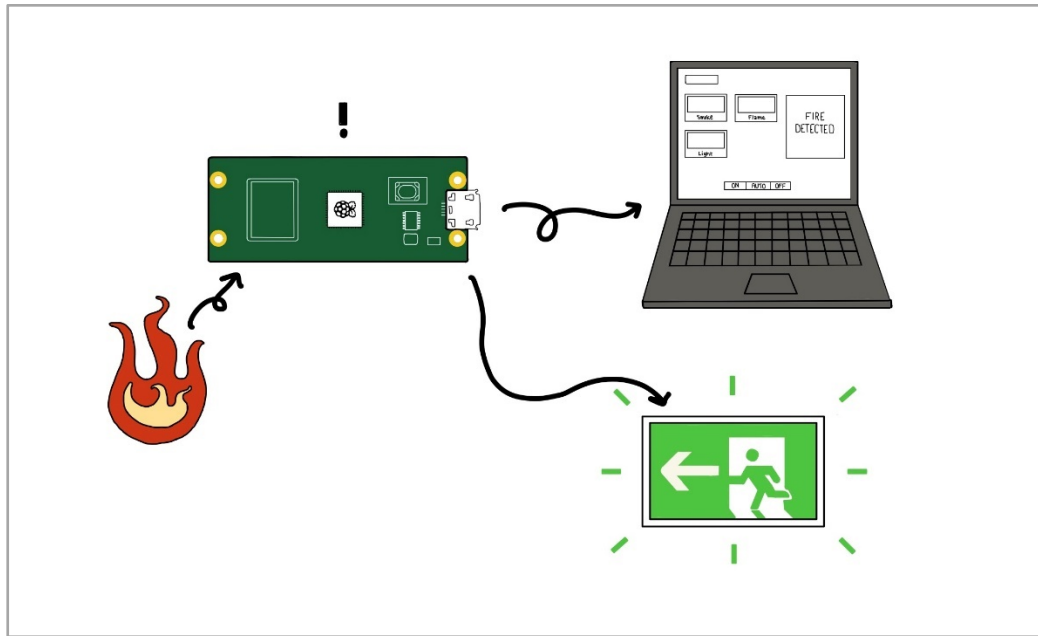
The following subsections describe each proposed solution in detail, including design justification, MVP technical description, and an initial implementation and beta testing plan.

## 2.1 SmartSafe: Integrated Fire Safety

The University of Victoria’s current fire safety system is prone to false alarms, and the emergency lighting system is ineffective in smoky conditions. Each fire alarm occurrence requires campus security officers and the Saanich and Oak Bay Fire departments to visit the location [12]. This causes unnecessary cost, stress, and danger for all parties involved. Furthermore, the current placement of the emergency exit signage on the ceiling decreases its visibility as smoke may block it, and occupants must crawl in the case of heavy smoke.

### 2.1.1 SmartSafe Concept Overview

The SmartSafe design proposes a three-sensor fire alarm system to increase the reliability of fire detection and decrease the frequency of false alarms, as well as low to the ground emergency lights to facilitate navigation in smoky conditions. The lights and alarm trigger automatically when at least two of the three sensors surpass their threshold values; however, a simple website informs the user if any of the three sensors reach their threshold value. **Figure 2** illustrates the general design concept.



**Figure 2. SmartSafe Full-Scale Design Concept**

Cheon et al. tested a similar design in “A Single-Chip CMOS Smoke and Temperature Sensor for an Intelligent Fire Detector” [13]. They found smoke sensors provide a quick response time, but they lacked reliability. Sulthana et al. investigated the strengths and weaknesses of various sensors in “Review Study on Recent Developments in Fire Sensing Methods” [14]. They emphasize that no sensor can perfectly detect fires with accuracy and speed. To mitigate this issue, our design uses two additional sensors to ensure consistent fire detection in the case of the smoke sensors’ failure or false alarm.

Chinchero, Alonso, and Ortiz explain in “LED lighting systems for smart buildings: a review” [15] that LED lights minimize the environmental impact of the design through low energy consumption and the use of sustainable materials, which supports UVic’s sustainability goals.

### 2.1.2 SmartSafe Proof-of-Concept

The system demonstrates the integration of multiple sensors to detect fires and help building occupants in emergency evacuations. The system uses a Raspberry Pi Pico W connected to a smoke, flame, and temperature sensor which constantly monitor the environment. If at least two out of the three sensors detect a fire, the Raspberry Pi Pico W prints a notification to the UI, and the LED lights turn on to mark a path to the nearest exit.

Each of the sensors monitors a different indicator of fire: this provides the system with multiple points of analysis in case one sensor malfunctions. If only one sensor surpasses its threshold value, the system waits until a second sensor is triggered before enabling the lighting system to minimize false alarms. The prototype demonstrates the utility and scalability of the SmartSafe concept to improve fire safety in UVic buildings.

The prototype implements the following sensors:

- **Smoke:** The MQ-135 gas sensor measures various gases in the air, such as smoke.
- **Flame:** The flame sensor module measures light waves to detect a flame and turns on its own LED if a flame is detected.
- **Temperature:** The DS18B20 sensor measures the temperature of the environment.

Refer to **Appendix C** for detailed circuit diagrams.

LED indicators will activate based on the following conditions:

- Flame:
  - ON: Flame detected near sensor
  - OFF: No flame detected
- Lighting:
  - ON: At least 2 sensors > threshold values
  - OFF: At least 2 sensors < threshold values

Refer to **Appendix D** for SmartSafe's documented code.

## Components and Breadboard Design

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**Figure 3** shows the MVP circuit design with three sensors connected and integrated using MicroPython code. **Appendix E** includes a full table of the parts and connections used in our circuits.

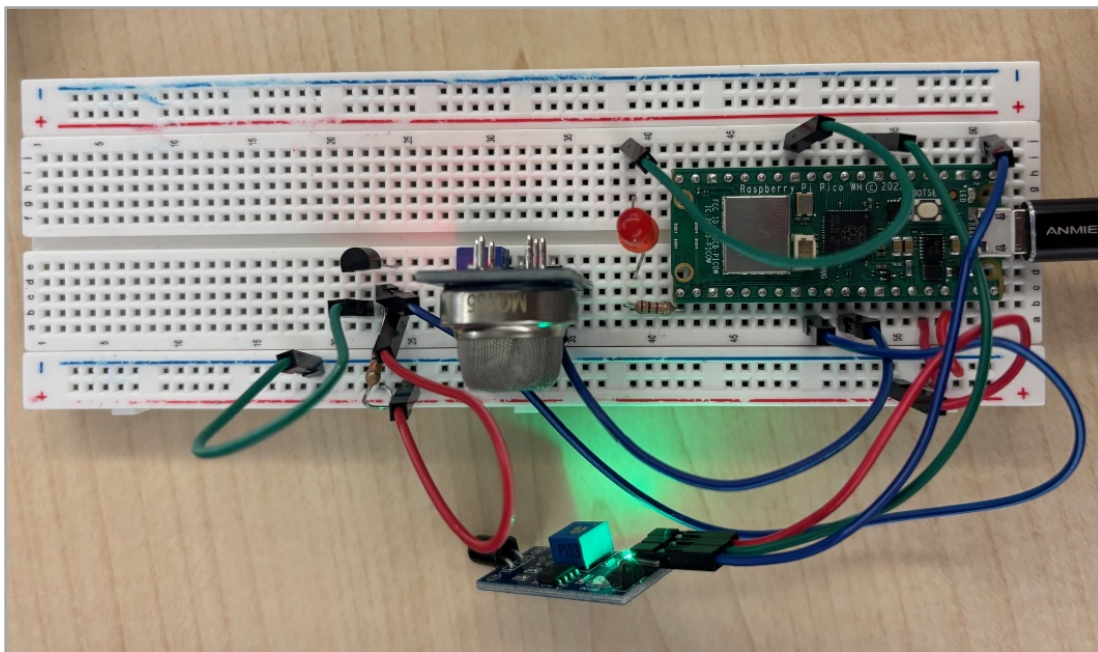


Figure 3. SmartSafe MVP Breadboard Prototype

**Table 3** describes the quantity and function of each component in SmartSafe’s design.

**Table 3. SmartSafe Components**

| Component                  | Quantity  | Function                                 |
|----------------------------|-----------|------------------------------------------|
| Raspberry Pi Pico W        | 1         | Processes sensor inputs and outputs      |
| Breadboard                 | 1         | Demonstrates circuit prototype           |
| MQ-135 Gas Sensor          | 1         | Detects smoke and airborne gasses        |
| Flame Sensor Module        | 1         | Detects infrared light produced by flame |
| DS18B20 Temperature Sensor | 1         | Measures ambient air temperature         |
| LED                        | 1         | Indicates emergency evacuation lighting  |
| Resistors                  | 2         | Limits current to LED’s                  |
| Wire                       | 12        | Connects circuit components              |
| <b>Total # Components</b>  | <b>20</b> |                                          |

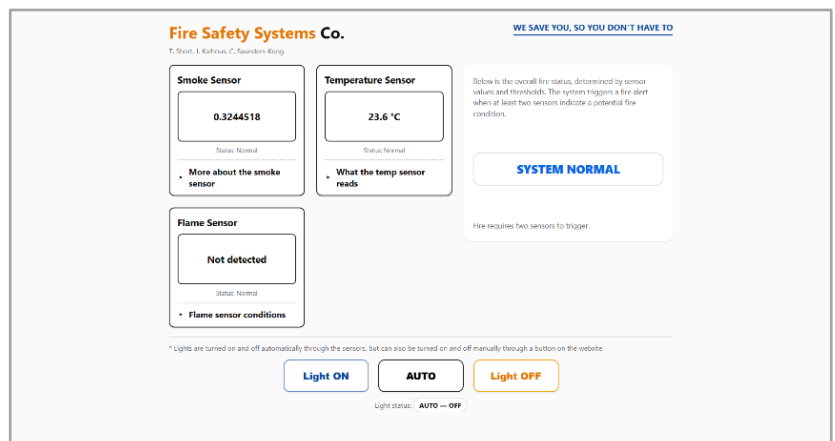
**Appendix F** outlines the cost for each sensor component from the SmartSafe design.

## SmartSafe UI

The user interface allows the user to access the sensor data and manually activate or deactivate the lighting system from a remote location in case the system fails to warn the user. **Figure 4** illustrates the UI design prototype, and **Figure 5** shows our working UI.



**Figure 4. SmartSafe UI Prototype**



**Figure 5. SmartSafe UI**

For an outline of how the SmartSafe UI satisfies the Universal Design Principles, refer to **Appendix G**.

### 2.1.3 SmartSafe Initial Implementation and Beta Testing Plan

---

The team will test the minimum viable product within one building on the UVic campus and place two units containing the three sensors and two emergency lights to ensure all parts are functional and properly integrated. The team will monitor the UI over the span of a week to test its functionality.

Beta testing includes the following:

- Check accuracy of temperature sensors over one week in relation to existing temperature sensors
- If a fire occurs, check that all three sensors surpass the determined threshold
- Run fire evacuation simulations and collect feedback on lighting accessibility

The data accumulated after the testing period allows for further troubleshooting and function improvement. The team will consider any necessary adjustments after testing and before campus-wide implementation, including unforeseen wiring or code errors, calibration of sensors, and range of reliable sensing.

## 2.2 SmartClimate: HVAC Optimization

---

Many buildings across the University of Victoria's campus feature outdated HVAC systems. As a result, high temperatures within their classrooms become uncomfortable. This poses an issue, threatening to distract students from academics [16].

For instance, the Clearihue building has a history of ongoing issues with its heating system. The University of Victoria's integrated energy masterplan (2011) outlines that Clearihue's heating system (gas fired boiler plant model) demonstrates the highest [17]:

- Number of connection points to existing heating loop (4)
- Loop connection cost (\$20,000)
- Amount of supplementary domestic hot water (DHW) generators (4)
- Supplementary DHW generator cost (\$40,000)

It also performs at the lowest energy output of 87,500 kWh/yr. Thus, the heating system implemented proves itself not only uncomfortable for students, but also inefficient.

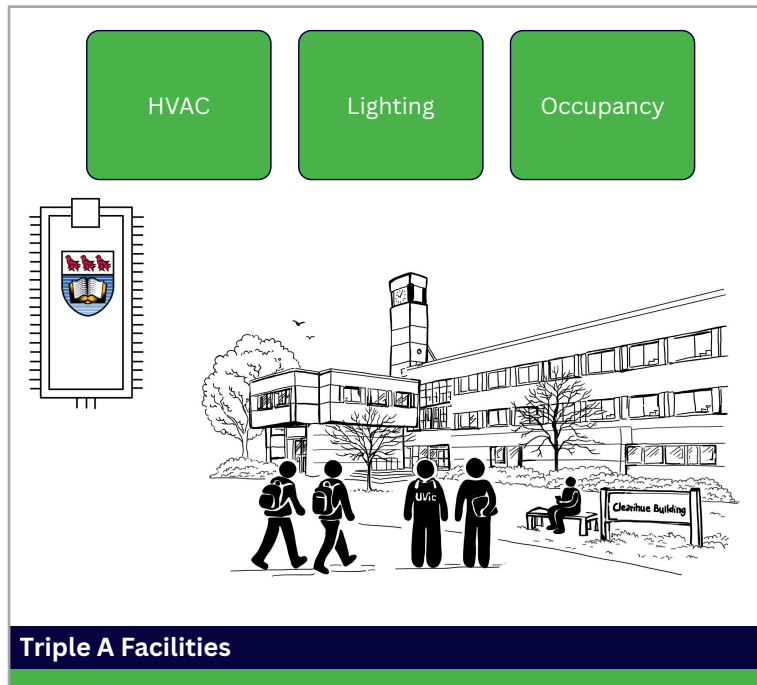
### 2.2.1 SmartClimate Design Overview

---

The SmartClimate MVP design addresses UVic's ineffective HVAC systems by monitoring and adjusting room temperature. It operates through a system of light, temperature, and occupancy sensors. Temperature sensing constantly acts to ensure classroom climate is within the range of comfortability. The other sensors act as supportive mitigation; if high occupancy is detected (through motion or light), the HVAC system is also turned on or up as a preventative measure. Implementing an approach of

automatic thermal and spatial sensing ensures an increased likelihood of classroom climate satisfaction in an efficient manner [18].

**Figure 6** demonstrates a full-scale design concept of SmartClimate.



**Figure 6. SmartClimate Full-Scale Design Concept**

In their experimental study, Aftab et al. proved that the implementation of an occupancy-based HVAC system, like **Figure 6**, increases comfort and efficiency [19].

### 2.2.2 SmartClimate Proof-of-Concept

The SmartClimate MVP strives for classroom comfortability, with occupants and efficiency at its core. It does so by implementing symbolic LED lights that operate by external influence and manual control to demonstrate building control.

Data gathered from various sensors update the user via the UI, to allow for classroom climate controls if plausible. Although the Raspberry Pi Pico functions on a smaller scale than buildings, it demonstrates its effectiveness on a larger system through simulation.

#### Sensors, Actuators, and System Control

The prototype implements three sensing systems:

- **Thermistor:** Senses temperature and adjusts resistance accordingly
- **Ambient Light Sensor:** Senses light and measures brightness
- **PIR Motion Sensor:** Detects motion by sensing changes in infrared radiation

Please refer to **Appendix H** for further detailed information on the stated sensing circuits.

LED indicators:

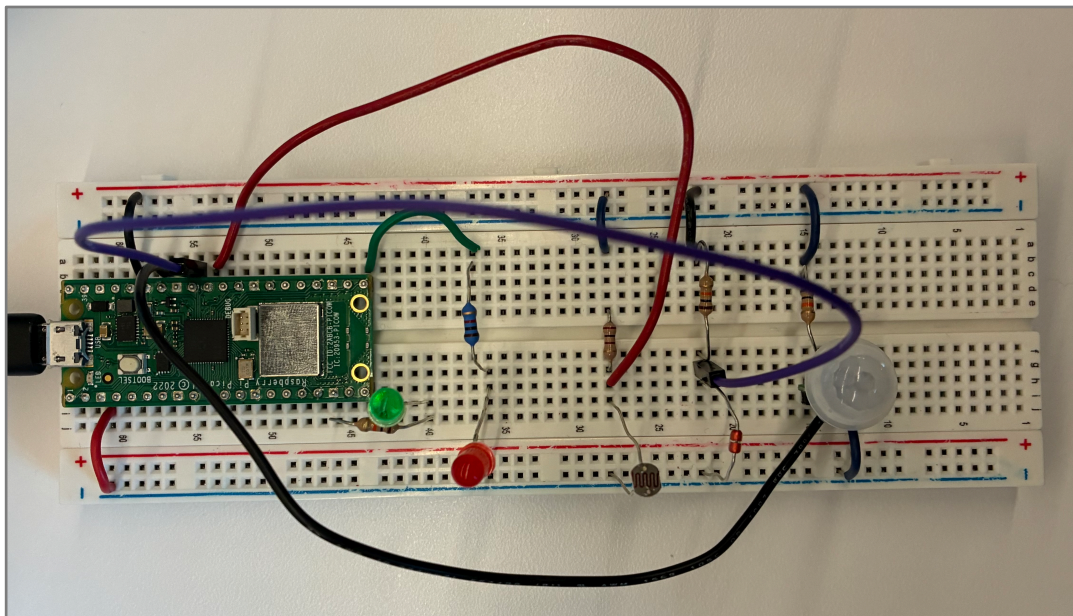
- Green LED:
  - ON: Room Temperature or Occupancy % > Threshold
    - AC turns on
  - OFF: Room Temperature or Occupancy % < Threshold
    - AC turns off
- Red LED:
  - ON: Classroom Light > Threshold
    - PIR motion sensor assessed
  - OFF: Classroom Light < Threshold
    - PIR motion sensor left untapped, thermistor continues to monitor for out-of-range temperatures

Refer to **Appendix I** for SmartClimate's documented code.

---

#### Components and Breadboard Design

**Figure 7** presents the SmartClimate MVP prototype. For a detailed connection mapping schematic, refer to **Appendix J**.



**Figure 7.** SmartClimate MVP Breadboard Prototype

**Table 4** outlines the design components and their function.

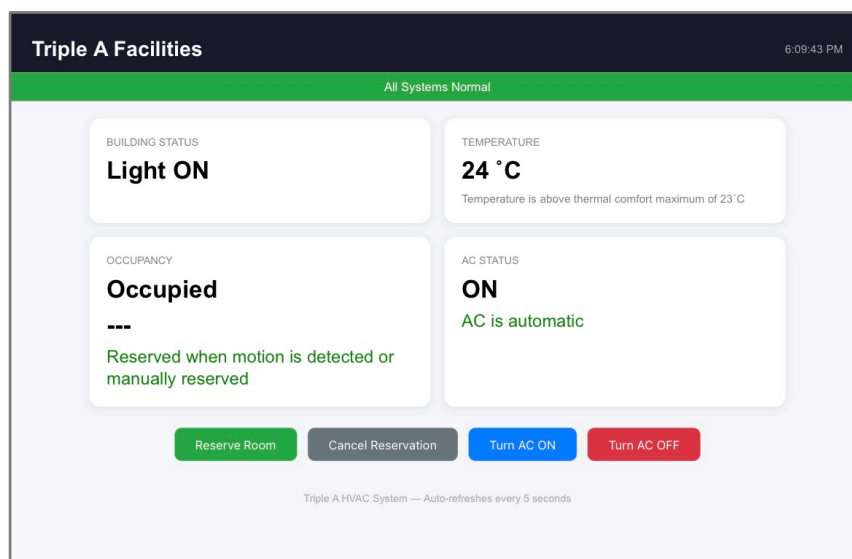
**Table 4. SmartClimate Components**

| Component                    | Quantity  | Function                                                                                                                       |
|------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------|
| NTC Thermistor               | 1         | Senses temperature and adjusts resistance accordingly.                                                                         |
| Ambient Light Sensor         | 1         | Senses light and measures brightness.                                                                                          |
| PIR Motion Sensor            | 1         | Detects motion by sensing changes in infrared radiation.                                                                       |
| Green LED                    | 1         | Visually indicates the behavior of the AC system as on or off. The brighter the light demonstrates a longer duration of AC on. |
| Red LED                      | 1         | Visually indicates the room occupancy. ON for occupants present, OFF for no occupants.                                         |
| Resistor (100 $\Omega$ )     | 3         | Protects LEDs by limiting current.                                                                                             |
| Resistor (10 k $\Omega$ )    | 1         | Protects sensors by limiting current.                                                                                          |
| Wire                         | 9         | Connects components to allow current flow through the breadboard.                                                              |
| <b>Total # of Components</b> | <b>18</b> |                                                                                                                                |

For an estimated per-zone sensor cost, please refer to **Appendix K**.

#### SmartClimate UI

**Figure 8** exhibits the UI, which gives the user the ability to monitor and control the HVAC systems within academic buildings.



**Figure 8. SmartClimate UI**

The simple design ensures minimal user confusion by presenting information in a direct, succinct manner. It allows for two-way interaction by permitting manual override on the AC system at the click of a button. For a detailed accessibility outline, please refer to **Appendix L**.

### 2.2.3 SmartClimate Initial Implementation and Beta Testing Plan

---

Initial implementation of SmartClimate will focus on a single older campus building, such as Clearihue. Installing and connecting light, temperature, and motion sensors to a central controller will further evaluate system performance.

Beta testing includes the following:

- Testing the system over a period of at least 5 weekdays
- Collecting data based on weather predictions and time-of-day to understand further external impact
- Comparing original system vs. SmartClimate energy consumption

This data provides feedback on what further developments in older UVic buildings are required for full-scale implementation.

## 2.3 SmartZone: Integrated Occupancy-Based HVAC and Lighting

---

HVAC and lighting systems across UVic's campus operate in isolation, relying on fixed schedules and showing little awareness of real-time occupancy. As a result, empty spaces remain lit and conditioned, wasting energy and preventing central monitoring for system optimization.

UVic's current climate sustainability action plan commits to significantly reducing operational emissions by 2030 [20]. The District Energy Plant, UVic's largest source of greenhouse gas (GHG) emissions [21], powers most campus heating systems. Additionally, heating and lighting represent the two largest sources of institutional energy consumption in Canada [22]. To meet sustainability targets, UVic therefore needs a system that offers improved efficiency.

### 2.3.1 SmartZone Concept Overview

---

The SmartZone MVP addresses this by providing a central monitoring system that integrates occupancy-aware HVAC and lighting systems to reduce unnecessary energy consumption. The MVP operates within a single building zone in McPherson Library, monitoring temperature, ambient light, and occupancy conditions while displaying data in real-time on the wireless UI. HVAC and lighting systems within the zone respond dynamically to sensor-observed conditions and manual control inputs via the UI. **Figure 9** illustrates the SmartZone concept in action.

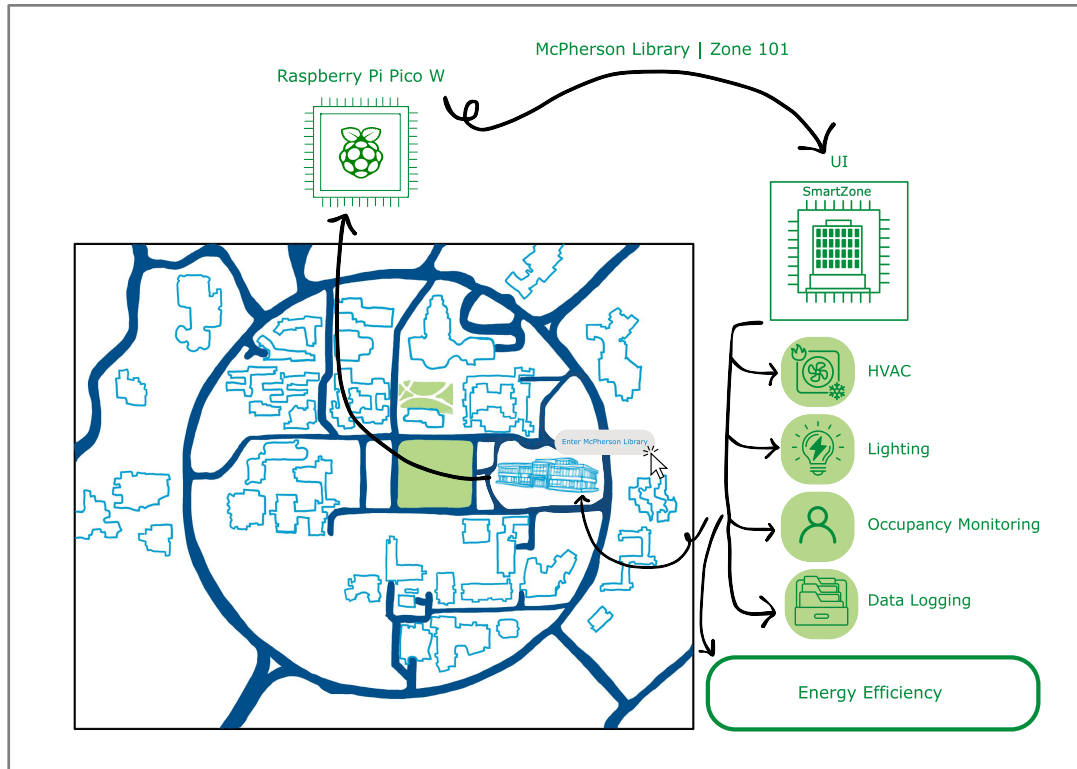


Figure 9. SmartZone Full-Scale Design Concept

Similar systems that implement occupancy-based control on Raspberry Pi hardware exhibit significant energy-efficiency increases [23], [34], supporting the feasibility and potential impact of this concept.

### 2.3.2 SmartZone Proof-of-Concept

The Raspberry Pi Pico W-based MVP demonstrates SmartZone’s function on a zone-representative breadboard. This validates the ability to scale to an integrated BMS that offers accessible control and energy-optimized operation.

The microcontroller receives and processes sensor inputs, hosting and updating the UI in real-time, and executing commands to turn on indicators. The accurate control of LEDs demonstrates how the design will provide occupancy-aware control of HVAC and lighting systems when scaled.

#### Sensors, Actuators, and System Control

The prototype implements three sensing systems:

- **Occupancy:** An IR emitter/receiver pair detects entry into the zone
- **Ambient Light:** A photoresistor measures available daylight
- **Temperature:** An NTC thermistor monitors zone temperature

Please refer to **Appendix M** for detailed technical explanations of each circuit.

LED indicators activate according to the following zone conditions:

- Lighting (Green):
  - ON: Occupied & Daylight < Threshold
  - OFF: Unoccupied or Daylight > Threshold
- Heating (Red):
  - ON: Occupied & Temperature < Heating Threshold
  - OFF: Unoccupied or Temperature > Heating Threshold
- Cooling (Blue):
  - ON: Occupied & Temperature > Cooling Threshold
  - OFF: Unoccupied or Temperature < Cooling Threshold

The SmartZone program runs on a structured file system. When activated, it prints prompts to the console to guide the user through calibration, setting adjustment, and different operational modes. The software also includes a preliminary log.py function to demonstrate long-term monitoring and system optimization potential.

A public repository contains the code and its documentation. **Appendix N** provides a link and a comprehensive summary of its contents.

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#### Components and Breadboard Design

**Figure 10** displays an image of the MVP prototype. For a detailed connection mapping schematic, please refer to **Appendix O**.

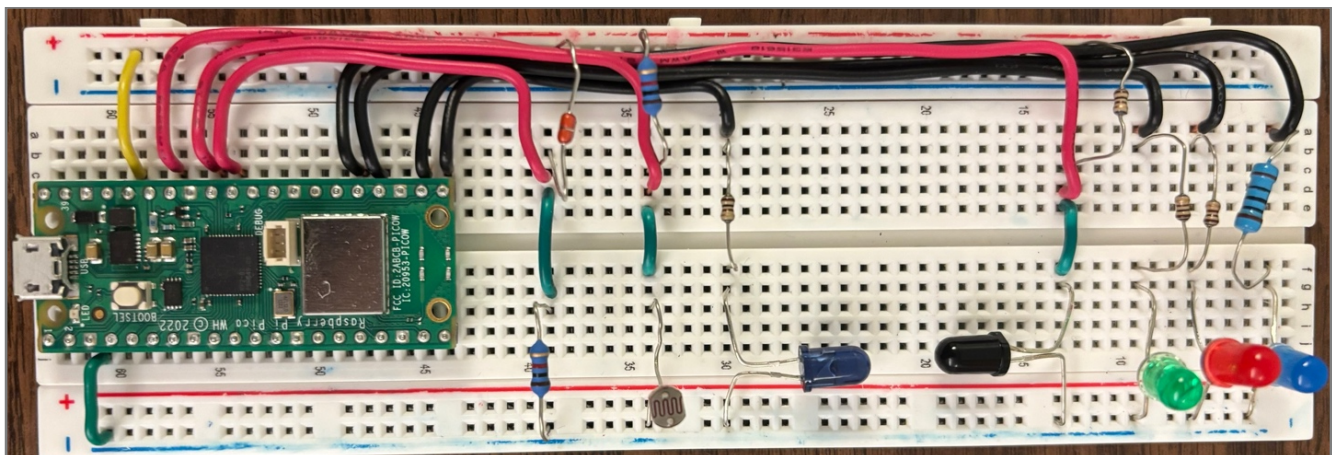


Figure 10. SmartZone MVP Breadboard Prototype

**Table 5** outlines the design components and their function.

**Table 5. SmartZone Components**

| Component                 | Quantity  | Function                                                                                                                    |
|---------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------|
| Breadboard                | 1         | Enables component electrical connection.                                                                                    |
| Raspberry Pi Pico W       | 1         | Provides central control by receiving and processing sensor inputs, hosting the wireless UI, and executing output commands. |
| NTC Thermistor            | 1         | Measures ambient temperature for heating and cooling control using voltage divider.                                         |
| CDS Photoresistor         | 1         | Measures ambient light level to enable daylight responsive lighting control.                                                |
| IR Phototransistor        | 1         | Detects infrared beam interruption to determine occupancy state.                                                            |
| IR LED                    | 1         | Emits infrared signal for occupancy detection.                                                                              |
| Green LED                 | 1         | Indicates lighting actuator activation state.                                                                               |
| Red LED                   | 1         | Indicates heating actuator activation state.                                                                                |
| Blue LED [25]             | 1         | Indicates cooling actuator activation state.                                                                                |
| Resistor (100 $\Omega$ )  | 5         | Limits current to protect LEDs and IR emitter.                                                                              |
| Resistor (10 k $\Omega$ ) | 2         | Provides voltage divider for input signals.                                                                                 |
| Wire                      | 12        | Connects components on the breadboard.                                                                                      |
| <b>Total # Components</b> | <b>28</b> |                                                                                                                             |

All parts additional to the project kit came from existing materials at no extra cost.

For an estimated per-zone sensor cost, please refer to **Appendix P**.

#### SmartZone UI

The UI provides a central monitoring interface and illustrates feasible provision to buildings across UVic's campus. The UI supports accessibility through a dyslexia-friendly font (Verdana) and clear visual navigation. For a detailed outline of how SmartZone implements the 7 Principles of Universal Design, please see **Appendix Q**.

**Figure 11** provides a visual mock-up of the SmartZone UI.

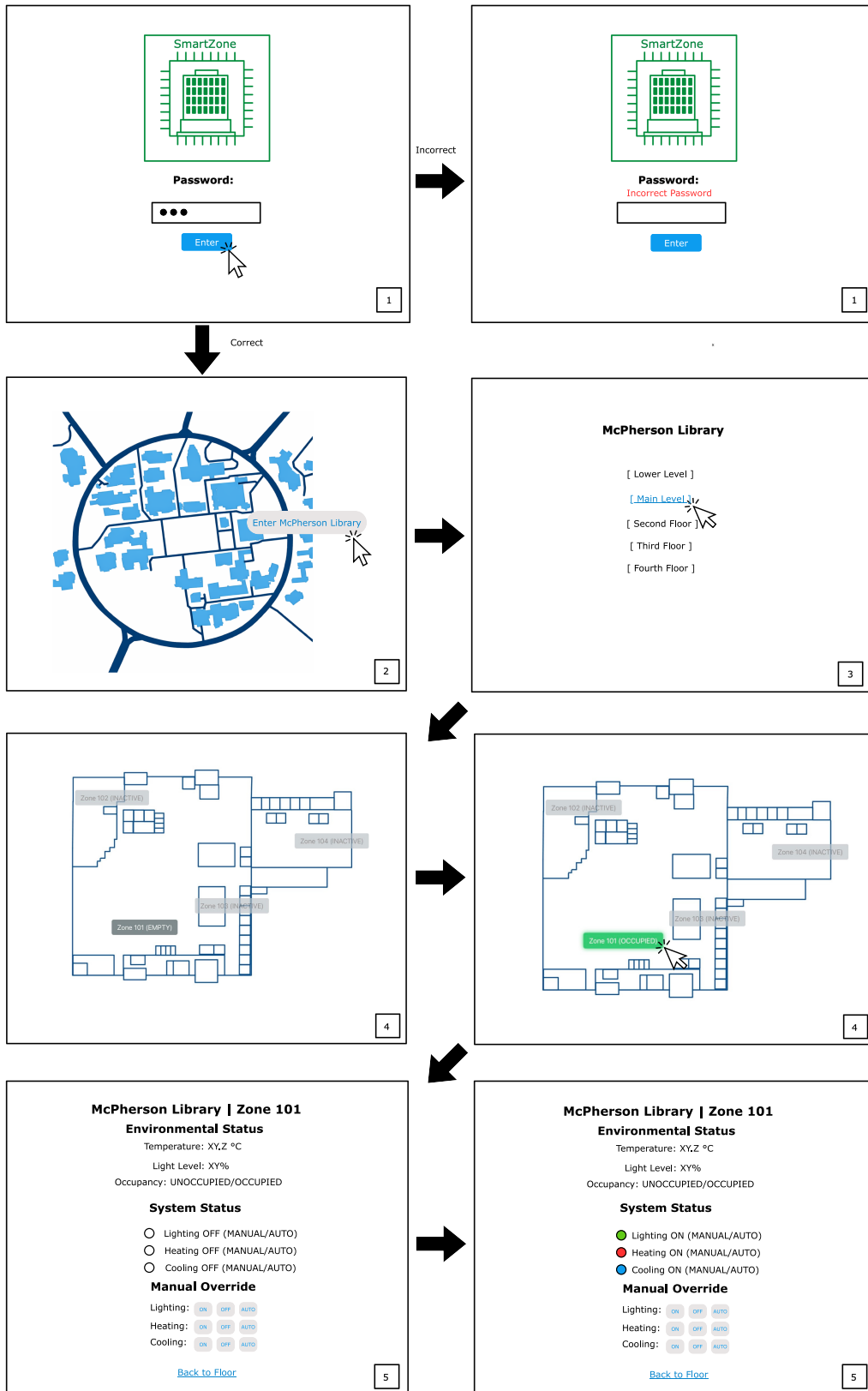


Figure 11. SmartZone UI

SmartZone protects against unauthorized access through dual password protection: a unique password is needed for both network connection and interface access. The map of Ring Road demonstrates the feasibility of being scaled to provide campus-wide monitoring at UVic, while the McPherson Library implementation shows building-scale representation. Manual overrides provide fail-safe measures and enable two-way user-UI interaction.

### 2.3.3 SmartZone Initial Implementation and Beta Testing Plan

To further evaluate performance, SmartZone will pilot within a single campus zone in the McPherson Library. Implementation includes installing temperature, daylight, and occupancy sensors and connecting HVAC and lighting actuators to the central controller.

Beta testing includes

- Monitoring system response accuracy over multiple occupancy cycles
- Logging temperature and light data over a 7-day period
- Comparing baseline energy consumption to that achieved with occupancy-aware control
- Collecting Facilities Management feedback on UI usability

Results provide data on the required system calibrations and refinements, along with the actuator upgrades needed for full-scale implementation.

## 2.4 Design Performance Results

This section outlines the laboratory performance of each prototype design and their projected energy savings based on supporting research.

### 2.4.1 Laboratory Testing Results

**Table 6** details how SmartSafe, SmartClimate, and SmartZone performed during primary testing.

**Table 6. Prototype Performance Results**

| Objective            | Metric | SmartSafe | SmartClimate | SmartZone |
|----------------------|--------|-----------|--------------|-----------|
| Real-Time Visibility | %      | 80%       | 70%          | 50%       |
| Condition Response   | %      | 100%      | 80%          | 93.3%     |
| System Integration   | #      | 2         | 2            | 3         |

For the detailed testing data, refer to **Appendix R** (SmartSafe), **Appendix S** (SmartClimate), and **Appendix T** (SmartZone).

### 2.4.2 Energy Efficiency Quantification

To quantify the energy-saving potential of each design, the team conducted secondary research. The estimated building savings is calculated as (% of Building Energy Use) x (% Potential Savings). **Table 7** provides a summary of each system's estimated efficiency.

**Table 7. Energy Efficiency Estimation Table**

| Design              | Function                        | % of Building Energy Use [26] | % Potential Savings | % Estimated Building Savings |
|---------------------|---------------------------------|-------------------------------|---------------------|------------------------------|
| <b>SmartSafe</b>    | Emergency Lighting + Fire Alarm | < 15.1 [27]                   | 30 [28]             | < 4.5                        |
| <b>SmartClimate</b> | Heating (HVAC)                  | 56.6                          | 17-24 [29]          | 9.6-13.6                     |
| <b>SmartZone</b>    | Lighting                        | 14.2                          | 20-60 [30]          | 13.3-23.2                    |
|                     | Heating + Cooling (HVAC)        | 61.1                          | 17-24               |                              |

## 2.5 Comparative Analysis of Proposed Prototypes

The team compared the three proposed designs based on how effectively each meets the objectives outlined in **Table 1**. We assigned scores to each objective according to its relative relevance, then evaluated each design using predefined 5-point scoring metrics derived from laboratory testing and secondary research. Please refer to **Appendix U** for detailed scoring tables and **Appendix V** for an extensive outline of each objective’s measurement and ranking rationale.

The team weighted accessibility the highest because RFP-202601 explicitly defines UVic’s need for a user-friendly system; the BuildHers aim to deliver a design that provides the best functionality for the greatest diversity of users.

**Table 8** provides an evaluation of each design across all objectives.

**Table 8. BuildHers Design Evaluation Chart**

| Objective             | Wt.  | Meas. Param         | SmartSafe |       |       | SmartClimate |       |       | SmartZone  |       |       |
|-----------------------|------|---------------------|-----------|-------|-------|--------------|-------|-------|------------|-------|-------|
|                       |      |                     | Magnitude | Score | Value | Magnitude    | Score | Value | Magnitude  | Score | Value |
| Accessibility         | 7/28 | # UD Principles     | 6         | 4     | 1     | 4            | 3     | 3/4   | 6          | 4     | 1     |
| System Integration    | 6/28 | # Systems           | 2         | 3     | 9/14  | 2            | 3     | 9/14  | 3          | 4     | 6/7   |
| Scalability           | 5/28 | \$ CAD              | \$17.24   | 2     | 5/14  | \$3.92       | 4     | 5/7   | \$3.01     | 4     | 5/7   |
| Real-Time Visibility  | 4/28 | % Timely Updates    | 80%       | 4     | 4/7   | 70%          | 3     | 3/7   | 50%        | 2     | 2/7   |
| Condition Response    | 3/28 | % Correct Responses | 100%      | 5     | 15/28 | 80%          | 4     | 3/7   | 93.3%      | 4     | 3/7   |
| Energy Efficiency     | 2/28 | % Savings           | < 4.5%    | 2     | 1/7   | 9.6-13.6%    | 4     | 2/7   | 13.3-23.2% | 5     | 5/14  |
| System Longevity      | 1/28 | # Parts             | 20        | 3     | 3/28  | 18           | 3     | 3/28  | 28         | 2     | 1/14  |
| Overall Utility Score |      |                     |           |       | 47/14 |              |       | 47/14 |            |       | 52/14 |

The comparative analysis highlights each design's strengths and trade-offs. These results inform the evaluation summary and final recommendation in the following sections.

### 3. Design Evaluation Summary

The comparative analysis shows that each design addresses a distinct aspect of UVic's building management needs, but none independently satisfies all RFP-202601 objectives.

SmartSafe has a high scaling cost and limited system integration but delivers strong accessibility and reliable safety responses. SmartClimate exhibits weaker accessibility and less consistent condition response accuracy, yet presents a low-cost, straightforward approach to improving thermal comfort. SmartZone shows lower real-time UI performance and system longevity, while achieving the greatest overall score, with high system integration and potential energy savings.

Regarding implementation, SmartClimate offers the simplest initial deployment, while SmartZone provides the most viable foundation for scalable, integrated building control. SmartSafe contributes essential safety functionality; however, it requires combination with other systems to meet broader operational objectives.

Primary testing results and supporting research therefore indicate that SmartZone most effectively meets the defined criteria overall, but refinement is necessary.

### 4. Final Design Recommendation

Based on the design evaluation and RFP-202601 objectives, the BuildHers recommend that UVic Facilities Management pursue the development of an integrated BMS design that builds on SmartZone's framework while incorporating key features from SmartSafe and SmartClimate.

SmartZone provides a strong foundation for integration and energy efficiency but requires improvement in safety functionality and UI simplicity. Implementing SmartSafe's multi-sensor fire detection and SmartClimate's straightforward temperature control will optimize the design for both user and occupant experience. We recommend further iteration and laboratory testing of the integrated prototype to refine system responsiveness and ensure the feasibility of scaling for deployment across campus.

This combined design offers UVic a practical means of centralized, accessible smart management. It has the potential to improve safety and enhance comfort for building occupants, as well as reduce operational energy use through an intuitive interface.

The BuildHers welcome your feedback on this recommendation and request approval to proceed with the further development and testing of this integrated solution.

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## Appendices

This section provides supplementary information.

### Appendix A: Team Contribution and Responsibility Logs

To ensure efficient task completion and support equitable teamwork, we utilized detailed schedule tracking sheets.

**Table 9** presents an overview of the report work scheduling information.

**Table 9. The BuildHers Work Log**

| Date Assigned                                  | Due    | Task Description                    | Assigned to    | Date Completed | Total Time Spent (Hrs) | Edited By | Date Revisions Completed | Signed Off By | Date of Sign Off |
|------------------------------------------------|--------|-------------------------------------|----------------|----------------|------------------------|-----------|--------------------------|---------------|------------------|
| <b>Front Matter</b>                            |        |                                     |                |                |                        |           |                          |               |                  |
| 27-Feb                                         | 20-Mar | Letter of Transmittal               | Claire         | 20-Mar         | 1.00                   | Sabrina   | 27-Mar                   | All           | 30-Mar           |
| 27-Feb                                         | 08-Mar | Title Page                          | Ayla           | 07-Mar         | 1.00                   | Claire    | 08-Mar                   | All           | 08-Mar           |
| 27-Feb                                         | 08-Mar | Table of Contents                   | Claire         | 07-Mar         | 0.50                   | Sabrina   | 08-Mar                   | All           | 08-Mar           |
| 27-Feb                                         | 12-Mar | List of Figures and Tables          | Sabrina        | 07-Mar         | 0.25                   | Ayla      | 08-Mar                   | All           | 08-Mar           |
| 27-Feb                                         | 20-Mar | Executive Summary                   | Claire         | 20-Mar         | 1.50                   | Julia     | 27-Mar                   | All           | 30-Mar           |
| 27-Feb                                         | 20-Mar | Glossary                            | All            | 07-Mar         | 0.50                   | All       | 08-Mar                   | Sabrina       | 08-Mar           |
| <b>Problem Definition/Introductory Section</b> |        |                                     |                |                |                        |           |                          |               |                  |
| 27-Feb                                         | 08-Mar | Client Background                   | Julia          | 07-Mar         | 1.25                   | Claire    | 08-Mar                   | All           | 08-Mar           |
| 27-Feb                                         | 08-Mar | Problem Definition                  | Julia          | 07-Mar         | 1.25                   | Claire    | 08-Mar                   | All           | 08-Mar           |
| 02-Mar                                         | 08-Mar | BuildHers Approach                  | Julia          | 09-Mar         | 1.5                    | Sabrina   | 10-Mar                   | All           | 10-Mar           |
| 02-Mar                                         | 08-Mar | Envisioned Solution Figure          | Sabrina        | 05-Mar         | 0.25                   | All       | 06-Mar                   | Julia         | 06-Mar           |
| <b>Proposed Solutions</b>                      |        |                                     |                |                |                        |           |                          |               |                  |
| 02-Mar                                         | 08-Mar | Solution Section Overview           | Sabrina        | 07-Mar         | 1.00                   | Claire    | 08-Mar                   | All           | 08-Mar           |
| 02-Mar                                         | 08-Mar | SmartSafe Description               | Julia & Claire | 12-Mar         | 11.00                  | Ayla      | 13-Mar                   | Ayla          | 13-Mar           |
| 02-Mar                                         | 08-Mar | SmartClimate Description            | Ayla           | 12-Mar         | 9.00                   | Julia     | 13-Mar                   | Julia         | 13-Mar           |
| 02-Mar                                         | 08-Mar | SmartZone Description               | Sabrina        | 07-Mar         | 10.00                  | Claire    | 08-Mar                   | Claire        | 08-Mar           |
| 02-Mar                                         | 08-Mar | Potential Energy Quantification     | Ayla           | 08-Mar         | 1.25                   | Julia     | 08-Mar                   | Julia         | 08-Mar           |
| <b>Comparative Analysis</b>                    |        |                                     |                |                |                        |           |                          |               |                  |
| 27-Feb                                         | 08-Mar | Comparative Analysis Overview       | Sabrina        | 05-Mar         | 0.75                   | Julia     | 06-Mar                   | Julia         | 06-Mar           |
| 02-Mar                                         | 20-Mar | Weighted Objectives Chart           | All            | 18-Mar         | 0.75                   | All       | 20-Mar                   | Sabrina       | 20-Mar           |
| <b>Summarizing Sections</b>                    |        |                                     |                |                |                        |           |                          |               |                  |
| 27-Feb                                         | 20-Mar | Conclusion                          | Sabrina        | 20-Mar         | 1.25                   | Claire    | 27-Mar                   |               | 27-Mar           |
| 27-Feb                                         | 20-Mar | Recommendation                      | Sabrina        | 20-Mar         | 1.00                   | Claire    | 27-Mar                   |               | 27-Mar           |
| 27-Feb                                         | 08-Mar | References (IEEE)                   | All            | 12-Mar         | 1.75                   | All       | 13-Mar                   | Sabrina       | 13-Mar           |
| <b>Back Matter</b>                             |        |                                     |                |                |                        |           |                          |               |                  |
| 27-Feb                                         | 27-Mar | Work Logs                           | Sabrina        | 27-Mar         | 1.50                   | Claire    | 31-Mar                   | Claire        | 01-Apr           |
| 02-Mar                                         | 08-Mar | Primary Research Procedure          | Ayla           | 07-Mar         | 0.25                   | Julia     | 08-Mar                   | Julia         | 08-Mar           |
| 10-Mar                                         | 27-Mar | SmartSafe Sensing Circuits          | Julia          | 20-Mar         | 1.50                   | Sabrina   | 20-Mar                   | Sabrina       | 27-Mar           |
| 10-Mar                                         | 27-Mar | SmartSafe Code Documentation        | Claire         | 20-Mar         | 0.75                   | Sabrina   | 20-Mar                   | Sabrina       | 27-Mar           |
| 10-Mar                                         | 27-Mar | SmartSafe Breadboard Diagram        | Julia          | 20-Mar         | 0.50                   | Sabrina   | 20-Mar                   | Sabrina       | 27-Mar           |
| 02-Mar                                         | 13-Mar | SmartSafe Estimated Sensor Costs    | Julia          | 12-Mar         | 0.50                   | Claire    | 13-Mar                   | Claire        | 13-Mar           |
| 02-Mar                                         | 13-Mar | SmartSafe UD Principles             | Julia          | 12-Mar         | 0.75                   | Claire    | 13-Mar                   | Claire        | 13-Mar           |
| 02-Mar                                         | 27-Mar | SmartSafe Testing Data              | Claire         | 25-Mar         | 2.00                   | Sabrina   | 27-Mar                   | Sabrina       | 27-Mar           |
| 10-Mar                                         | 27-Mar | SmartClimate Circuits               | Ayla           | 26-Mar         | 1.50                   | Claire    | 27-Mar                   | Claire        | 27-Mar           |
| 10-Mar                                         | 27-Mar | SmartClimate Code Documentation     | Ayla           | 27-Mar         | 0.50                   | Julia     | 27-Mar                   | Julia         | 27-Mar           |
| 10-Mar                                         | 27-Mar | SmartClimate BreadBoard Diagram     | Ayla           | 27-Mar         | 0.75                   | Claire    | 27-Mar                   | Claire        | 27-Mar           |
| 02-Mar                                         | 13-Mar | SmartClimate Estimated Sensor Costs | Ayla           | 12-Mar         | 0.75                   | Sabrina   | 13-Mar                   | Sabrina       | 13-Mar           |
| 02-Mar                                         | 13-Mar | SmartClimate UD Principles          | Ayla           | 12-Mar         | 0.75                   | Claire    | 13-Mar                   | Claire        | 13-Mar           |
| 02-Mar                                         | 27-Mar | SmartClimate Testing Data           | Ayla           | 27-Mar         | 2.50                   | Julia     | 27-Mar                   | Julia         | 27-Mar           |
| 10-Mar                                         | 27-Mar | SmartZone Circuits                  | Sabrina        | 07-Mar         | 1.50                   | Ayla      | 08-Mar                   | Ayla          | 08-Mar           |
| 10-Mar                                         | 27-Mar | SmartZone Code Documentation        | Sabrina        | 10-Mar         | 0.50                   | Ayla      | 10-Mar                   | Ayla          | 10-Mar           |
| 10-Mar                                         | 27-Mar | SmartZone BreadBoard Diagram        | Sabrina        | 10-Mar         | 2.00                   | Ayla      | 10-Mar                   | Ayla          | 10-Mar           |
| 02-Mar                                         | 13-Mar | SmartZone Estimated Sensor Costs    | Sabrina        | 10-Mar         | 0.50                   | Julia     | 13-Mar                   | Julia         | 13-Mar           |
| 02-Mar                                         | 13-Mar | SmartZone UD Principles             | Sabrina        | 10-Mar         | 0.50                   | Julia     | 13-Mar                   | Julia         | 13-Mar           |
| 02-Mar                                         | 27-Mar | SmartZone Testing Data              | Sabrina        | 08-Mar         | 2.50                   | Ayla      | 10-Mar                   | Ayla          | 10-Mar           |
| 02-Mar                                         | 05-Mar | Objectives Scoring Tables           | Ayla & Julia   | 06-Mar         | 1.50                   | Claire    | 08-Mar                   | Claire        | 08-Mar           |
| 02-Mar                                         | 05-Mar | Objective Ranking Rationale         | All            | 06-Mar         | 1.00                   | All       | 09-Mar                   | Sabrina       | 09-Mar           |

As two members of the BuildHers team (Julia and Claire) worked on one design concept (SmartSafe), they divided design description tasks equitably. **Table 10** illustrates the task breakdown.

**Table 10. SmartSafe Design Task Breakdown**

| Date Assigned                                         | Due    | Task Description                     | Assigned to    | Date Completed | Total Time Spent (Hrs) |
|-------------------------------------------------------|--------|--------------------------------------|----------------|----------------|------------------------|
| <b>Introduction Section</b>                           |        |                                      |                |                |                        |
| 02-Mar                                                | 08-Mar | Need Statement & Supporting Data     | Claire         | 08-Mar         | 1.00                   |
| 02-Mar                                                | 08-Mar | Concept MVP Overview                 | Claire         | 08-Mar         | 1.00                   |
| 02-Mar                                                | 08-Mar | Full-Scale Diagram                   | Claire         | 08-Mar         | 1.25                   |
| 02-Mar                                                | 08-Mar | Peer Reviewed Supporting Sources     | Claire & Julia | 08-Mar         | 1.00                   |
| <b>Technical Description</b>                          |        |                                      |                |                |                        |
| 02-Mar                                                | 08-Mar | Proof-of-Concept Overview            | Julia          | 08-Mar         | 1.00                   |
| 02-Mar                                                | 08-Mar | Sensors, Actuators, & System Control | Julia          | 08-Mar         | 1.00                   |
| 10-Mar                                                | 27-Mar | Components and Breadboard Design     | Julia          | 20-Mar         | 1.00                   |
| 02-Mar                                                | 27-Mar | Photo of Built Breadboard Design     | Claire         | 20-Mar         | 0.50                   |
| 02-Mar                                                | 08-Mar | Components Table                     | Julia          | 08-Mar         | 1.00                   |
| 02-Mar                                                | 08-Mar | UI Description & Drawing             | Claire         | 08-Mar         | 1.25                   |
| <b>Initial Implementation &amp; Beta Testing Plan</b> |        |                                      |                |                |                        |
| 02-Mar                                                | 08-Mar | Initial Testing: What, How, & Why    | Julia          | 08-Mar         | 1.00                   |
| <b>Appendices</b>                                     |        |                                      |                |                |                        |
| 10-Mar                                                | 27-Mar | Sensing Circuit 1                    | Julia          | 20-Mar         | 0.50                   |
| 10-Mar                                                | 27-Mar | Sensing Circuit 2                    | Julia          | 20-Mar         | 0.50                   |
| 10-Mar                                                | 27-Mar | Sensing Circuit 3                    | Julia          | 20-Mar         | 0.50                   |
| 10-Mar                                                | 27-Mar | Code Documentation (+ GitHub Link)   | Claire         | 20-Mar         | 0.75                   |
| 10-Mar                                                | 27-Mar | Breadboard Diagram + Component Map   | Julia          | 20-Mar         | 0.50                   |
| 02-Mar                                                | 08-Mar | UD Evaluation Table                  | Julia          | 12-Mar         | 0.75                   |
| 02-Mar                                                | 08-Mar | Estimated Sensor Cost                | Julia          | 12-Mar         | 0.50                   |
| 02-Mar                                                | 27-Mar | Primary Testing Data                 | Claire         | 25-Mar         | 2.00                   |
| <b>Formatted References</b>                           |        |                                      |                |                |                        |
| 02-Mar                                                | 08-Mar | Data Supporting Need                 | Claire         | 08-Mar         | 0.75                   |
| 02-Mar                                                | 08-Mar | Peer Reviewed Article 1              | Claire         | 08-Mar         | 0.50                   |
| 02-Mar                                                | 08-Mar | Peer Reviewed Article 2              | Julia          | 08-Mar         | 0.50                   |
| 02-Mar                                                | 13-Mar | Sensor Costs                         | Julia          | 12-Mar         | 0.50                   |
| 02-Mar                                                | 08-Mar | Potential Energy Savings             | Julia          | 08-Mar         | 0.50                   |

## Appendix B: Primary Research Procedure

The following standard laboratory testing procedure ensured comparable primary research:

- **Real-Time Visibility:** Record 10 trials, document whether the UI updated within the programmed refresh time on a pass/fail basis.
- **Condition Response Accuracy:** Record 10 trials per system, document if the indicators performed the expected response on a pass/fail basis and calculate the average accuracy.
- **Integrability:** Record the number of represented building systems providing functional data (>50% of responses accurate).

## Appendix C: SmartSafe Sensing Circuits

The SmartSafe design utilizes three sensor circuits to detect fire.

**Figure 12** illustrates the MQ-135 gas sensor circuit.

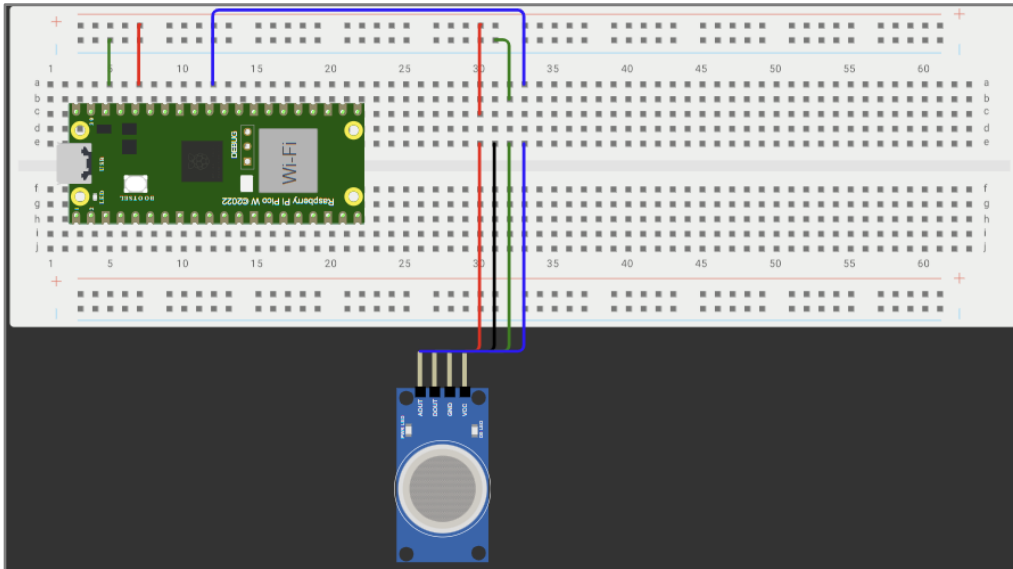


Figure 12. Gas Sensing Circuit

**Figure 13** shows the flame sensing module circuit.

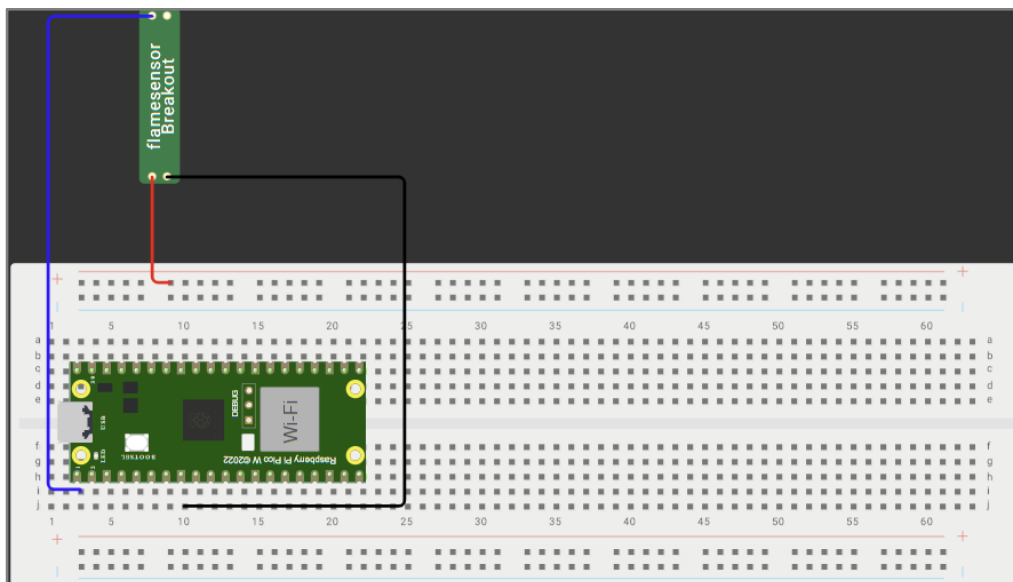
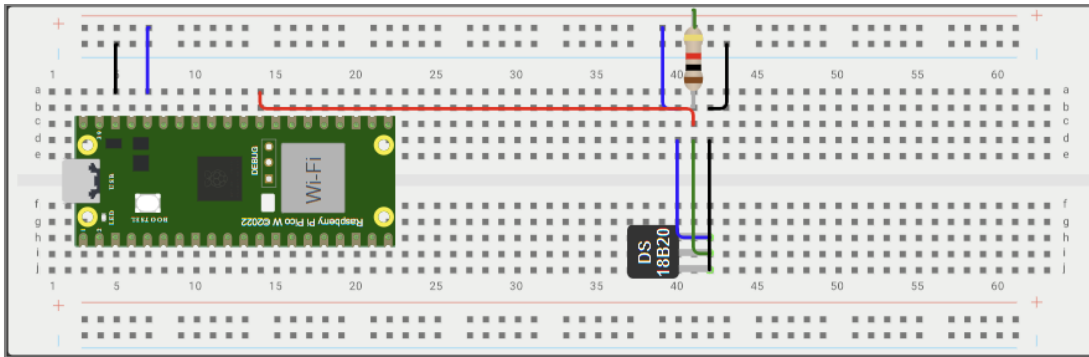


Figure 13. Flame Sensing Circuit

**Figure 14** displays the DS18B20 temperature sensor circuit.



**Figure 14. SmartSafe Temperature Circuit**

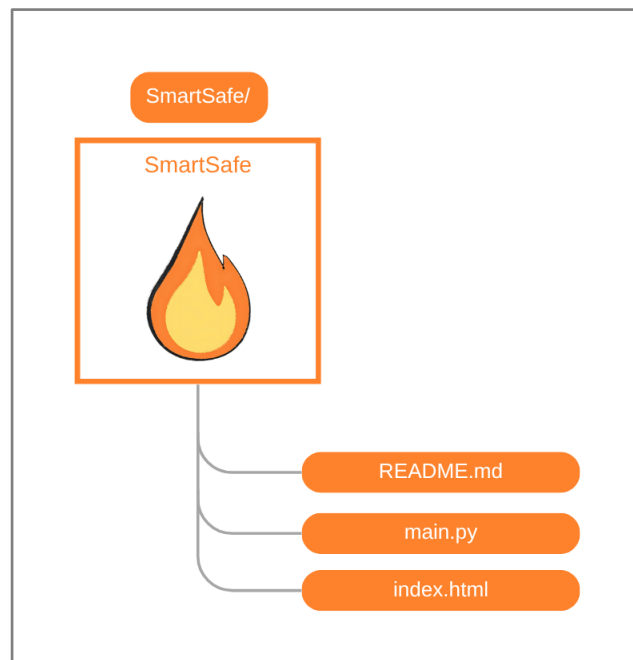
#### Appendix D: SmartSafe Software Documentation

The BuildHers GitHub repository provides the code written for the SmartSafe UI and Raspberry Pi Pico W.

Here is the link: <https://github.com/TheBuildHers/SmartSafe>

It contains the MicroPython and HTML code files needed to read and process the sensor values, turn on and off the LED, and run the SmartSafe webpage.

**Figure 15** illustrates the contents of the SmartSafe folder.



**Figure 15. SmartSafe Code Documentation Structure**

The SmartSafe folder contains the following files:

- **README.md**: explains the general function of the code
- **main.py**: stores the MicroPython code which runs the system
- **index.html**: contains the HTML code to maintain the SmartSafe webpage

## Appendix E: SmartSafe MVP Circuit Diagram

The SmartSafe MVP integrates three sensors and an LED actuator with the Raspberry Pi Pico W.

**Figure 16** shows a design concept of the MVP breadboard layout.

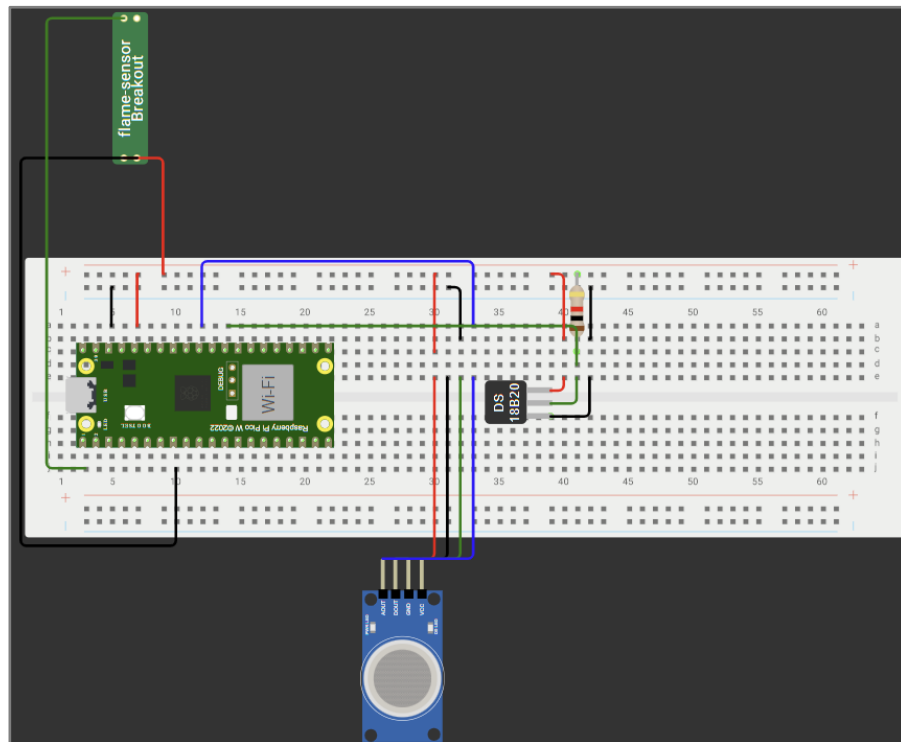


Figure 16. SmartSafe MVP Breadboard Layout

**Table 11** maps the parts and their connections.

Table 11. SmartSafe Component Connections

| Function           | Pin | GPIO/ADC | Type        |
|--------------------|-----|----------|-------------|
| Temperature Sensor | 22  | GPIO17   | Digital out |
| Smoke Sensor       | 12  | GPIO9    | Digital out |
| Flame sensor       | 1   | GPIO0    | Digital out |
| LED                | 25  | GPIO19   | Digital out |

## Appendix F: SmartSafe Estimated Per Zone Sensor Cost

**Table 12** provides an estimation of SmartSafe’s per-zone sensor costs to support scalability.

**Table 12. SmartSafe Estimated Per-Zone Sensor Cost**

| Sensor Type         | Function               | Part Number       | Cost (\$ CAD) | Reference Number |
|---------------------|------------------------|-------------------|---------------|------------------|
| Gas sensor          | Air quality monitoring | 1188-SNS-MQ135-ND | 6.89          | [31]             |
| Flame sensor module | Fire detection         | 1752              | 2.89          | [32]             |
| Digital thermometer | Temperature            | DS18B20           | 7.46          | [33]             |
| <b>Total Cost</b>   |                        |                   | <b>17.24</b>  |                  |

## Appendix G: SmartSafe’s Implementation of the 7 Principles of Universal Design

We outline how the SmartSafe user interface adheres to the 7 Principles of Universal Design in **Table 13**.

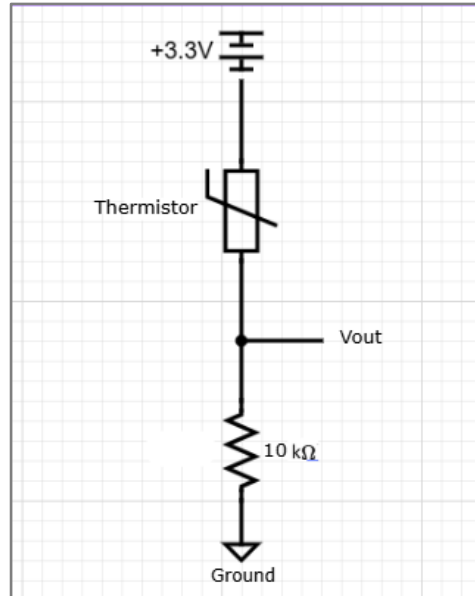
**Table 13. SmartSafe Universal Design Evaluation**

| #                                | Principle                           | Method of Implementation                                                                                                          | Satisfied  |
|----------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|------------|
| 1                                | Equitable use                       | Provides free access to the UI but limited to UVic FMGT to ensure security.                                                       | Yes        |
| 2                                | Flexibility in Use                  | Does not support different methods of use.                                                                                        | No         |
| 3                                | Simple and Intuitive Use            | Minimizes confusion and streamlines user experience through minimal design and visual hierarchy techniques.                       | Yes        |
| 4                                | Perceptible Information             | Explains the functions of each sensor using small boxes of information explaining the functions of each sensor.                   | Yes        |
| 5                                | Tolerance for Error                 | Bypasses user error by displaying a warning message to alert the user if any of the sensors reach their assigned threshold value. | Yes        |
| 6                                | Low Physical Effort                 | Minimizes physical effort as the user only needs to exert the force required to click a mouse and call emergency services.        | Yes        |
| 7                                | Size and Space for Approach and Use | Allows users to sit or stand while monitoring the UI and easily reach the mouse required to interact with the system.             | Yes        |
| <b>Final Accessibility Score</b> |                                     |                                                                                                                                   | <b>6/7</b> |

## Appendix H: SmartClimate Sensing Circuits

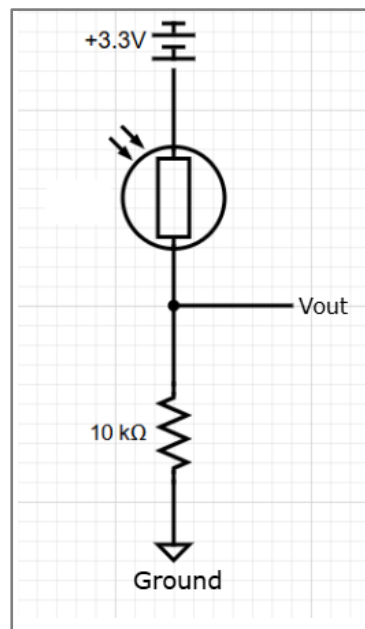
The SmartClimate prototype implements three sensing circuits.

**Figure 17** displays the thermistor circuit that performs temperature sensing.



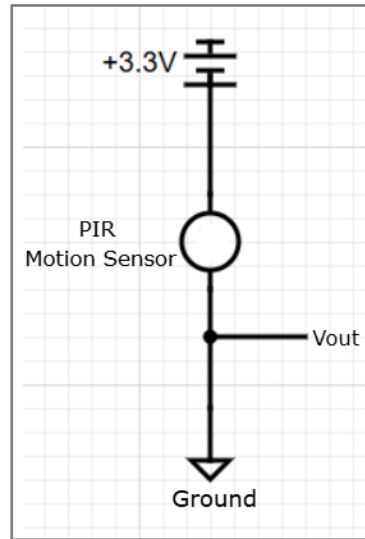
**Figure 17. SmartClimate Thermistor Circuit**

**Figure 18** shows the photoresistor circuit that provides light sensing functions.



**Figure 18. SmartClimate Photoresistor Circuit**

**Figure 19** displays a diagram of the PIR motion sensor that performs occupancy sensing.



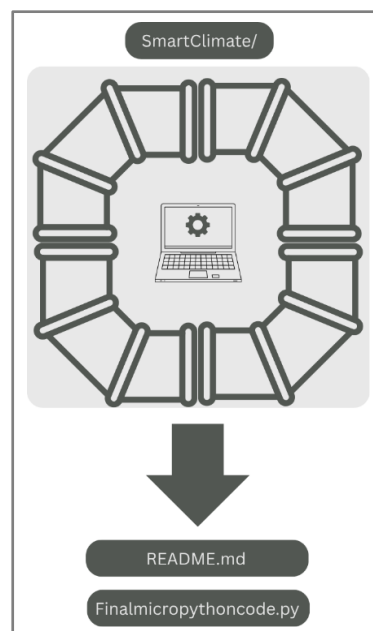
**Figure 19. SmartClimate PIR Motion Sensor Circuit**

#### Appendix I: SmartClimate Software Documentation

The BuildHers GitHub repository hosts the SmartClimate code:

<https://github.com/TheBuildHers/SmartClimate>

The software design is simple and intuitive; users only need to click run on Finalmicropythoncode.py to start the system. **Figure 20** illustrates the contents of the SmartClimate folder.



**Figure 20. SmartClimate File Structure Illustration**

## Appendix J: SmartClimate MVP Circuit Diagram

Figure 21 presents the SmartClimate MVP breadboard layout.

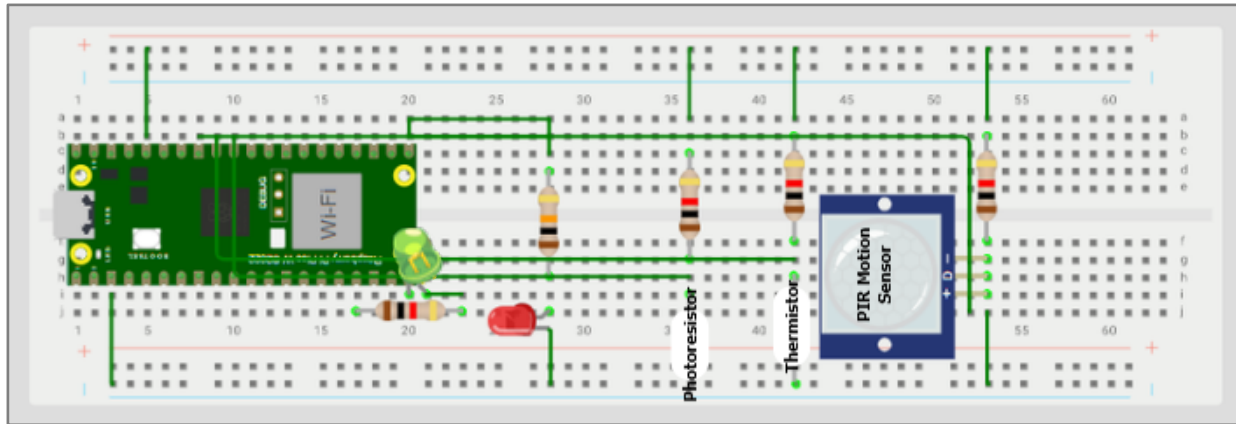


Figure 21. SmartClimate MVP Breadboard Layout

Table 17 summarizes the circuit connection mapping.

Table 14. SmartClimate Component Connections

| Function             | Pin | GPIO/ADC | Type        |
|----------------------|-----|----------|-------------|
| Ambient Light Sensor | 31  | ADC0     | Analog In   |
| Thermistor           | 32  | ADC1     | Analog In   |
| PIR Motion Detector  | 34  | GP28     | Analog In   |
| Green LED            | 20  | GP15     | Digital Out |
| Red LED              | 21  | GP16     | Digital Out |

## Appendix K: SmartClimate Estimated Per Zone Sensor Cost

Table 18 presents SmartClimate's expected cost per sensor.

Table 15. SmartClimate Estimated Per-Zone Sensor Cost

| Sensor Type         | Function    | Part Number | Cost (\$ CAD) | Reference Number |
|---------------------|-------------|-------------|---------------|------------------|
| NTC Thermistor      | Temperature | 490-8601-ND | 0.51          | [34]             |
| CdS Photoresistor   | Lighting    | 1528-161-ND | 1.46          | [35]             |
| PIR Motion Detector | Occupancy   | 528-5578-ND | 1.95          | [36]             |
| <b>Total Cost</b>   |             |             | <b>3.92</b>   |                  |

## Appendix L: SmartClimate's Implementation of the 7 Principles of Universal Design

This section outlines how the SmartClimate UI implements the 7 Principles of Universal Design.

**Table 19** displays SmartClimate's accessibility score.

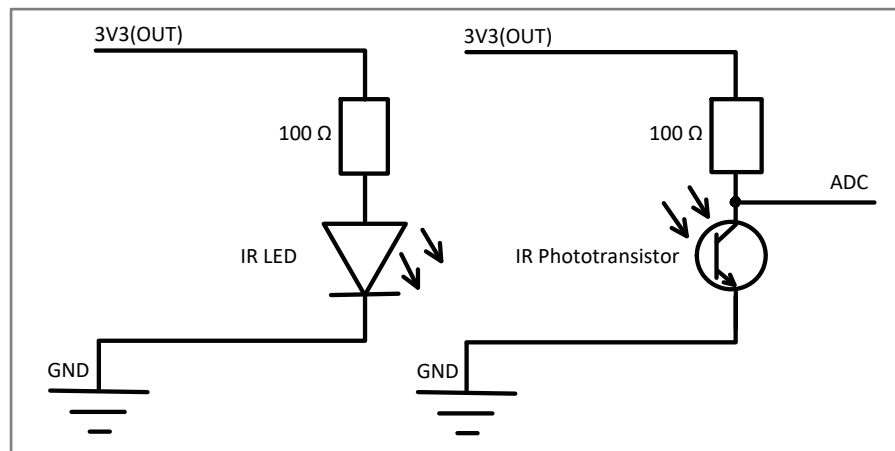
**Table 16. SmartClimate Universal Design Evaluation**

| #                                | Principle                           | Method of Implementation                                                                                    | Satisfied  |
|----------------------------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------|------------|
| 1                                | Equitable use                       | Uses real-time status to reduce uncertainty and allow for many ages and comprehension levels to understand. | Yes        |
| 2                                | Flexibility in Use                  | Does not implement flexibility in use.                                                                      | No         |
| 3                                | Simple and Intuitive Use            | Centralizes all controls and information to a single webpage for intuitive user experience.                 | Yes        |
| 4                                | Perceptible Information             | Uses simple, straightforward phrases and buttons.                                                           | Yes        |
| 5                                | Tolerance for Error                 | Does not implement tolerance for error measures.                                                            | No         |
| 6                                | Low Physical Effort                 | Provides only a few buttons that the user can/must interact with.                                           | Yes        |
| 7                                | Size and Space for Approach and Use | Does not implement size and space considerations.                                                           | No         |
| <b>Final Accessibility Score</b> |                                     |                                                                                                             | <b>4/7</b> |

## Appendix M: SmartZone Sensing Circuits

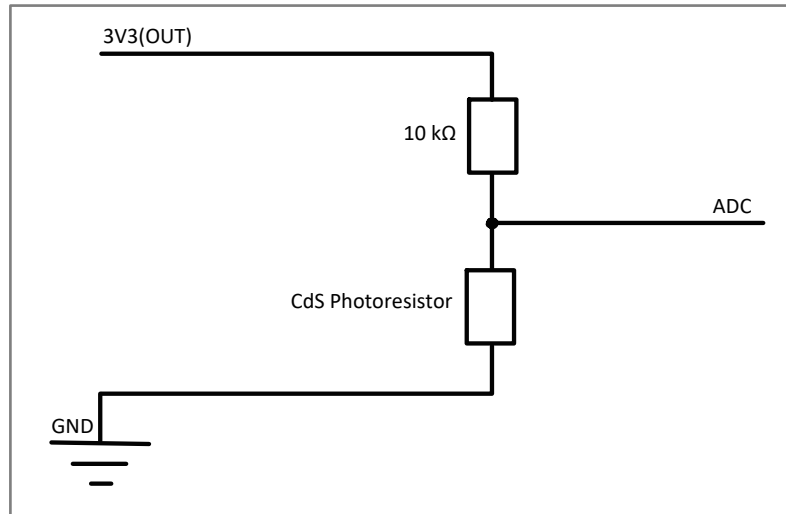
This section outlines the three sensing circuits that the SmartZone MVP uses.

SmartZone detects occupancy using an IR emitter (LED) and receiver (phototransistor) pair. When the IR beam breaks, the receiver's resistance drops, signalling occupancy detection. **Figure 22** shows a simplified circuit diagram.



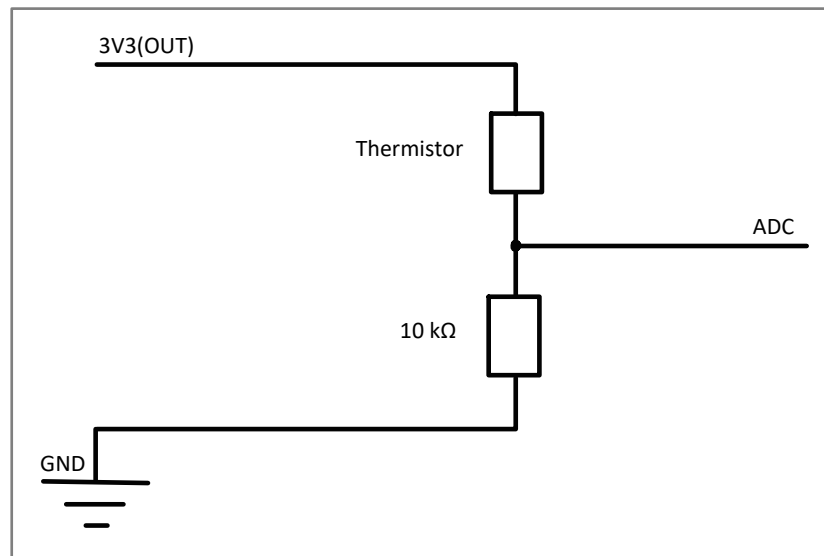
**Figure 22. IR Occupancy Circuit**

In the daylight sensing circuit, a phototransistor and a fixed 10 k $\Omega$  resistor form a voltage divider. Variations in daylight alter the photoresistor's resistance, which changes the divider voltage that the ADC input measures to calculate ambient light level. **Figure 23** illustrates this.



**Figure 23. Daylight Sensing Circuit**

The temperature sensing circuit uses an NTC thermistor in a voltage divider configuration with a fixed 10 k $\Omega$  resistor. **Figure 24** provides a simplified circuit schematic.



**Figure 24. Temperature Sensing Circuit**

When the temperature increases, the thermistor's resistance decreases, changing the voltage at the divider midpoint. The ADC input on the Raspberry Pi Pico W reads this voltage and converts it to temperature in degrees Celsius using the Thermistor's Beta equation.

## Appendix N: SmartZone Software Documentation

The complete SmartZone code and documentation are available at:  
<https://github.com/TheBuildHers/SmartZone>

The repository includes all code files used to process signal inputs, control the LEDs representing actuators, establish the network connection, and produce the UI. **Figure 25** represents the directory structure used.

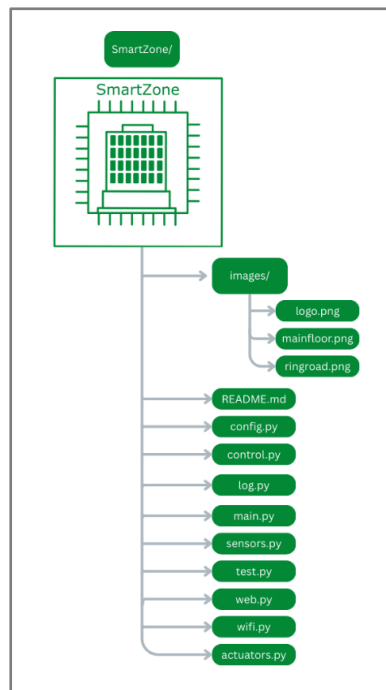


Figure 25. SmartZone File System Structure

The SmartZone folder contains the following:

- **Images/**
  - logo.png: SmartZone logo
  - mainfloor.png: McPherson Library mainfloor layout
  - ringroad.png: Campus map used in system UI
- **README.md**: Provides setup instructions and system overview
- **config.py**: Stores global variables and system settings
- **control.py**: Implements control algorithm to activate LED responses
- **log.py**: Enables data logging by writing system variables to a CSV file every 60 seconds
- **main.py**: Runs the SmartZone system
- **sensors.py**: Processes temperature, occupancy, and daylight inputs
- **test.py**: Allows the user to see the environmental conditions and set the configurations accordingly
- **wifi.py**: Handles network connection
- **actuators.py**: Activates LEDs to indicate system activation when called

## Appendix O: SmartZone MVP Circuit Diagram

Figure 26 illustrates the MVP breadboard connections.

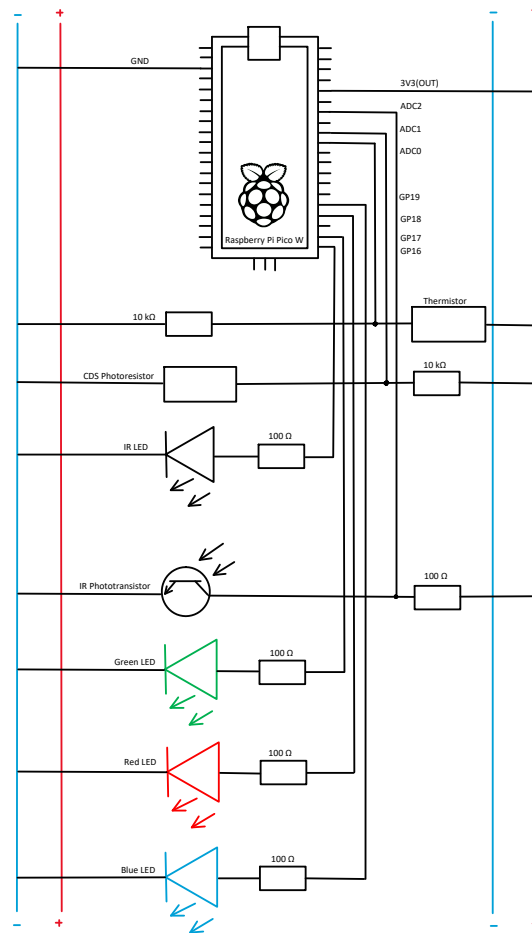


Figure 26. SmartZone MVP Breadboard Layout

Table 24 summarizes the circuit connection mapping.

Table 17. SmartZone Component Connections

| Function             | Pin | GPIO/ADC | Type        |
|----------------------|-----|----------|-------------|
| Thermistor           | 31  | ADC0     | Analog In   |
| Ambient Light Sensor | 32  | ADC1     | Analog In   |
| IR Phototransistor   | 34  | ADC2     | Analog In   |
| IR LED Emitter       | 21  | GP16     | Digital Out |
| Green LED            | 22  | GP17     | Digital Out |
| Red LED              | 24  | GP18     | Digital Out |
| Blue LED             | 25  | GP19     | Digital Out |

## Appendix P: SmartZone Estimated per Zone Sensor Cost

**Table 25** supports SmartZone’s scalability through secondary research on the design’s expected sensor cost per-zone.

**Table 18. SmartZone Estimated Per-Zone Sensor Cost**

| Sensor Type       | Function    | Part Number  | Cost (\$ CAD) | Reference Number |
|-------------------|-------------|--------------|---------------|------------------|
| NTC thermistor    | Temperature | 490-8601-ND  | 0.51          | [34]             |
| CdS photoresistor | Lighting    | 1528-161-ND  | 1.46          | [35]             |
| IR emitter        | Occupancy   | 1080-1080-ND | 0.49          | [37]             |
| IR receiver       |             | 160-1031-ND  | 0.55          | [38]             |
| Total Cost        |             |              | 3.01          |                  |

## Appendix Q: SmartZone’s Implementation of the 7 Principles of Universal Principles

**Table 26** outlines SmartZone’s accessibility score.

**Table 19. SmartZone Universal Design Evaluation**

| #                                | Principle                           | Method of Implementation                                                                                 | Satisfied  |
|----------------------------------|-------------------------------------|----------------------------------------------------------------------------------------------------------|------------|
| 1                                | Equitable use                       | Uses dyslexia friendly font (Verdana) to avoid stigmatizing users.                                       | Yes        |
| 2                                | Flexibility in Use                  | Provides user with the choice of passive monitoring or active control (visual display/manual overrides). | Yes        |
| 3                                | Simple and Intuitive Use            | Uses visual navigation (images of Ring Road and McPherson Library floor plan) to eliminate complexity.   | Yes        |
| 4                                | Perceptible Information             | Presents different modes of information via written and visual status indicators.                        | Yes        |
| 5                                | Tolerance for Error                 | Provides fail safe measures through manual override controls for each system.                            | Yes        |
| 6                                | Low Physical Effort                 | Eliminates physical interaction with automated control.                                                  | Yes        |
| 7                                | Size and Space for Approach and Use | Fails to implement spatial accessibility measures.                                                       | No         |
| <b>Final Accessibility Score</b> |                                     |                                                                                                          | <b>6/7</b> |

## Appendix R: SmartSafe Primary Testing Results

This section summarizes SmartSafe’s results from primary laboratory testing.

The code configures SmartSafe to update every 2 seconds. **Table 14** shows the refresh success rate achieved during MVP testing.

**Table 20. SmartSafe UI Refresh Success Rate Chart**

| <b>Trial #</b>   | <b>Update Time Taken (s)</b> | <b>Refresh Within Time? (Pass/Fail)</b> |
|------------------|------------------------------|-----------------------------------------|
| 1                | 1.94                         | P                                       |
| 2                | 1.79                         | P                                       |
| 3                | 2.03                         | F                                       |
| 4                | 1.99                         | P                                       |
| 5                | 1.83                         | P                                       |
| 6                | 1.91                         | P                                       |
| 7                | 2.18                         | F                                       |
| 8                | 1.96                         | P                                       |
| 9                | 1.99                         | P                                       |
| 10               | 2.00                         | P                                       |
| <b>Pass Rate</b> |                              | <b>80%</b>                              |

**Table 15** displays the emergency lighting and fire detection response system accuracy.

**Table 21. SmartSafe Lighting Response Accuracy**

| <b>System:</b> | <b>Lighting</b>                            |                          |                        |                  |
|----------------|--------------------------------------------|--------------------------|------------------------|------------------|
| <b>Trial #</b> | <b>Condition</b>                           | <b>Excepted Response</b> | <b>Actual Response</b> | <b>Pass/Fail</b> |
| 1              | Smoke and flame present                    | Lighting ON              | Lighting ON            | P                |
| 2              | Smoke and flame present                    | Lighting ON              | Lighting ON            | P                |
| 3              | Smoke and high temperature present         | Lighting ON              | Lighting ON            | P                |
| 4              | Smoke and high temperature present         | Lighting ON              | Lighting ON            | P                |
| 5              | Flame and high temperature present         | Lighting ON              | Lighting ON            | P                |
| 6              | Flame and high temperature present         | Lighting ON              | Lighting ON            | P                |
| 7              | High temperature present                   | Lighting OFF             | Lighting OFF           | P                |
| 8              | Flame present                              | Lighting OFF             | Lighting OFF           | P                |
| 9              | Smoke present                              | Lighting OFF             | Lighting OFF           | P                |
| 10             | Smoke, flame, and high temperature present | Lighting ON              | Lighting ON            | P                |

**Table 16** summarizes SmartSafe’s overall condition response accuracy and displays its number of functional systems.

**Table 22. SmartSafe System Accuracy**

| <b>System</b>                | <b>% Accuracy</b> | <b>Functional? (&gt;50%)</b> |
|------------------------------|-------------------|------------------------------|
| Fire detection               | 100%              | ✓                            |
| Lighting                     | 100%              | ✓                            |
| <b># Functional Systems:</b> | <b>100 %</b>      | <b>2</b>                     |

## Appendix S: SmartClimate Primary Testing Results

This section demonstrates SmartClimate's performance during testing in the laboratory.

**Table 20** shows the refresh success rate achieved.

**Table 23. SmartClimate UI Refresh Success Rate Chart**

| Trial #   | Update Time Taken | Refresh Within Time? (Pass/Fail) |
|-----------|-------------------|----------------------------------|
| 1         | 2.2               | P                                |
| 2         | 1.8               | P                                |
| 3         | 2.5               | F                                |
| 4         | 1.6               | P                                |
| 5         | 2.0               | F                                |
| 6         | 1.8               | P                                |
| 7         | 1.5               | P                                |
| 8         | 1.4               | P                                |
| 9         | 2.2               | F                                |
| 10        | 1.7               | P                                |
| Pass Rate |                   | 70%                              |

**Table 21** demonstrates the heating system results ( $T$  = temperature and  $TH$  = cooling threshold).

**Table 24. SmartClimate Heating Response Accuracy**

| System: | AC                      |                             |                 |           |
|---------|-------------------------|-----------------------------|-----------------|-----------|
| Trial # | Condition               | Excepted Response           | Actual Response | Pass/Fail |
| 1       | Occupied & $T < TH$     | Motion sensor (MS) assessed | MS assessed     | P         |
| 2       | Occupied & $T < TH$     | MS assessed                 | MS assessed     | P         |
| 3       | Occupied and $T > TH$   | AC On                       | On              | P         |
| 4       | Occupied and $T > TH$   | AC On                       | Off             | F         |
| 5       | Occupied and $T > TH$   | AC On                       | On              | P         |
| 6       | Occupied and $T = TH$   | AC On                       | Off             | F         |
| 7       | Occupied and $T = TH$   | AC On                       | On              | P         |
| 8       | Unoccupied and $T = TH$ | AC On                       | On              | P         |
| 9       | Unoccupied and $T < TH$ | AC Off                      | Off             | P         |
| 10      | Unoccupied and $T < TH$ | AC Off                      | Off             | P         |

**Table 22** displays the lighting system response accuracy.

**Table 25. SmartClimate Lighting Response Accuracy**

| <b>System:</b> | <b>Lighting</b>    |                          |                                      |                  |
|----------------|--------------------|--------------------------|--------------------------------------|------------------|
| <b>Trial #</b> | <b>Condition</b>   | <b>Excepted Response</b> | <b>Actual Response</b>               | <b>Pass/Fail</b> |
| 1              | Light Detected     | MS assessed              | MS assessed                          | P                |
| 2              | Light Detected     | MS assessed              | MS not assessed                      | F                |
| 3              | Light Detected     | Motion sensor assessed   | MS assessed                          | P                |
| 4              | Light Detected     | MS assessed              | MS assessed                          | P                |
| 5              | Light Detected     | MS assessed              | MS assessed                          | P                |
| 6              | Light not detected | System does not act      | Thermistor assessed, MS assessed     | F                |
| 7              | Light not detected | System does not act      | Thermistor assessed, MS not assessed | P                |
| 8              | Light not detected | System does not act      | Thermistor assessed, MS not assessed | P                |
| 9              | Light not detected | System does not act      | Thermistor assessed, MS not assessed | P                |
| 10             | Light not detected | System does not act      | Thermistor assessed, MS not assessed | P                |

**Table 23** summarizes SmartClimate's overall condition response accuracy.

**Table 26. SmartClimate System Accuracy**

| <b>System</b>                | <b>% Accuracy</b> | <b>Functional? (&gt;50%)</b> |
|------------------------------|-------------------|------------------------------|
| Cooling/AC                   | 80%               | ✓                            |
| Lighting                     | 80%               | ✓                            |
| <b># Functional Systems:</b> | <b>80%</b>        | <b>2</b>                     |

## Appendix T: SmartZone Primary Testing Results

This section provides an evaluation of SmartZone’s performance during final laboratory testing.

The code configures SmartZone to update every 1 second. **Table 27** summarizes the refresh success rate achieved during MVP testing.

**Table 27. SmartZone UI Refresh Success Rate Chart**

| Trial #   | Update Time Taken (s) | Refresh Within Time? (Pass/Fail) |
|-----------|-----------------------|----------------------------------|
| 1         | 2.4                   | F                                |
| 2         | 1.6                   | F                                |
| 3         | 2.3                   | F                                |
| 4         | 1.0                   | P                                |
| 5         | 1.0                   | P                                |
| 6         | 2.4                   | F                                |
| 7         | 1.0                   | P                                |
| 8         | 1.3                   | F                                |
| 9         | 1.0                   | P                                |
| 10        | 1.0                   | P                                |
| Pass Rate |                       | 50%                              |

The Raspberry Pi Pico W operates at near full memory capacity; the UI’s complex features come with the trade-off of a slow runtime.

The team conducted 10 trials per system and recorded LED responses to measure success. A variable potentiometer simulated temperature changes during testing. **Table 28** shows the heating system results ( $T$  = temperature,  $TL$  = heating threshold, and  $TH$  = cooling threshold).

**Table 28. SmartZone Heating Response Accuracy**

| System: | Heating                  |                   |                 |           |
|---------|--------------------------|-------------------|-----------------|-----------|
| Trial # | Condition                | Excepted Response | Actual Response | Pass/Fail |
| 1       | Occupied & $T > TH$      | Heating Off       | Off             | P         |
| 2       | Occupied & $T > TH$      | Heating Off       | Off             | P         |
| 3       | Occupied & $TL < T < TH$ | Heating Off       | Off             | P         |
| 4       | Occupied & $TL < T < TH$ | Heating Off       | Off             | P         |
| 5       | Occupied & $T < TL$      | Heating On        | On              | P         |
| 6       | Occupied & $T = TL$      | Heating Off       | Off             | P         |
| 7       | Occupied & $T = TH$      | Heating Off       | Off             | P         |
| 8       | Unoccupied & $T > TH$    | Heating Off       | Off             | P         |
| 9       | Unoccupied & $T < TL$    | Heating Off       | Off             | P         |
| 10      | Occupied & $TL < T < TH$ | Heating Off       | Off             | P         |

**Table 29** displays the cooling system response accuracy.

**Table 29. SmartZone Cooling Response Accuracy**

| <b>System:</b> | <b>AC (Cooling)</b>      |                          |                        |                  |
|----------------|--------------------------|--------------------------|------------------------|------------------|
| <b>Trial #</b> | <b>Condition</b>         | <b>Excepted Response</b> | <b>Actual Response</b> | <b>Pass/Fail</b> |
| 1              | Occupied & $T > TH$      | Cooling On               | On (delayed)           | F                |
| 2              | Occupied & $T > TH$      | Cooling On               | On                     | P                |
| 3              | Occupied & $TL < T < TH$ | Cooling Off              | Off                    | P                |
| 4              | Occupied & $TL < T < TH$ | Cooling Off              | Off                    | P                |
| 5              | Occupied & $T < TL$      | Cooling Off              | Off                    | P                |
| 6              | Occupied & $T = TL$      | Cooling Off              | Off                    | P                |
| 7              | Occupied & $T = TH$      | Cooling Off              | Off                    | P                |
| 8              | Unoccupied & $T > TH$    | Cooling Off              | Off                    | P                |
| 9              | Unoccupied & $T < TL$    | Cooling Off              | Off                    | P                |
| 10             | Occupied & $TL < T < TH$ | Cooling Off              | Off                    | P                |

**Table 30** presents SmartZone's lighting response accuracy ( $DL$  = measured daylight level,  $T$  = threshold % below which lighting activates).

**Table 30. SmartZone Lighting Response Accuracy**

| <b>System:</b> | <b>Lighting</b>       |                          |                        |                  |
|----------------|-----------------------|--------------------------|------------------------|------------------|
| <b>Trial #</b> | <b>Condition</b>      | <b>Excepted Response</b> | <b>Actual Response</b> | <b>Pass/Fail</b> |
| 1              | Occupied & $DL > T$   | Lighting Off             | Off                    | P                |
| 2              | Occupied & $DL > T$   | Lighting Off             | Off                    | P                |
| 3              | Occupied & $DL < T$   | Lighting On              | On                     | P                |
| 4              | Occupied & $DL < T$   | Lighting On              | On                     | P                |
| 5              | Occupied & $DL = T$   | Lighting Off             | Off                    | P                |
| 6              | Occupied & $DL = T$   | Lighting Off             | Off                    | P                |
| 7              | Unoccupied & $DL > T$ | Lighting Off             | Off                    | P                |
| 8              | Unoccupied & $DL > T$ | Lighting Off             | Off                    | P                |
| 9              | Unoccupied & $DL < T$ | Lighting Off             | Off (delayed change)   | F                |
| 10             | Unoccupied & $DL = T$ | Lighting Off             | Off                    | P                |

**Table 31** summarizes SmartZone’s overall condition response accuracy.

**Table 31. SmartZone System Accuracy**

| System                     | % Accuracy   | Functional? (>50%) |
|----------------------------|--------------|--------------------|
| Heating                    | 100%         | ✓                  |
| Cooling                    | 90%          | ✓                  |
| Lighting                   | 90%          | ✓                  |
| <b>Functional Systems:</b> | <b>93.3%</b> | <b>3</b>           |

## Appendix U: Objective Scoring Tables

This appendix provides the scoring tables corresponding to each objective within the Weighted Objectives Chart.

**Table 32** outlines the accessibility scoring metrics based on the number of Universal Design principles the MVP design implements (out of 7).

**Table 32. Scoring Table for Accessibility**

| Number of Universal Design Principles Implemented | Score |
|---------------------------------------------------|-------|
| 0 principles                                      | 1     |
| 1-2 principles                                    | 2     |
| 3-4 principles                                    | 3     |
| 5-6 principles                                    | 4     |
| All 7 principles                                  | 5     |

**Table 33** outlines the system integration scoring metrics based on the number of systems the MVP successfully demonstrates control of.

**Table 33. Scoring Table for System Integration**

| Number of Systems Integrated | Score |
|------------------------------|-------|
| 0 systems                    | 1     |
| 1 system                     | 2     |
| 2 systems                    | 3     |
| 3 systems                    | 4     |
| 4+ systems                   | 5     |

**Table 34** outlines the scalability scoring metrics based on the estimated per-zone sensor cost.

**Table 34. Scoring Table for Scalability**

| Estimated Per-Zone Cost (\$ CAD) | Score |
|----------------------------------|-------|
| > \$20.00                        | 1     |
| \$15.00-19.99                    | 2     |
| \$5.00-14.99                     | 3     |
| \$1.00-4.99                      | 4     |
| < \$1.00                         | 5     |

**Table 35** outlines the real-time visibility scoring metrics based on the percentage of successful UI refreshes measured during primary testing.

**Table 35. Scoring Table for Real-Time Visibility**

| Refresh Success Rate (%) | Score |
|--------------------------|-------|
| 0%                       | 1     |
| 1-49%                    | 2     |
| 50-79%                   | 3     |
| 80-99%                   | 4     |
| 100%                     | 5     |

**Table 36** outlines the condition response scoring metrics based on the percentage of correct system responses measured during primary testing.

**Table 36. Scoring Table for Condition Response**

| % Correct Responses | Score |
|---------------------|-------|
| < 50%               | 1     |
| 50-64%              | 2     |
| 65-79%              | 3     |
| 80-94%              | 4     |
| 95-100%             | 5     |

**Table 37** outlines the energy efficiency scoring metrics based on the estimated potential found through secondary research.

**Table 37. Scoring Table for Energy Efficiency**

| Projected Energy Reduction (%) | Score |
|--------------------------------|-------|
| 0%                             | 1     |
| 1-5%                           | 2     |
| 6-8%                           | 3     |
| 9-15%                          | 4     |
| >15%                           | 5     |

**Table 38** outlines the system longevity scoring metrics based on the number of components used in the MVP design. This represents system complexity and anticipates the maintenance needs for each design.

**Table 38. Scoring Table for System Longevity**

| Number of MVP Components | Score |
|--------------------------|-------|
| > 30                     | 1     |
| 26-30                    | 2     |
| 15-25                    | 3     |
| 10-15                    | 4     |
| < 10                     | 5     |

## Appendix V: Objective Ranking Rationale

**Table 39** displays the team’s pairwise evaluation of objectives. Weight calculations add one point to each score and normalize by the sum of the ranking series.

**Table 39. Pairwise Comparison of Objectives**

|                      | Accessibility | Integrability | Scalability | Real-Time Visibility | Condition Response | Energy Efficiency | System Longevity | Total |
|----------------------|---------------|---------------|-------------|----------------------|--------------------|-------------------|------------------|-------|
| Accessibility        | -             | 1             | 1           | 1                    | 1                  | 1                 | 1                | 6     |
| Integrability        | 0             | -             | 1           | 1                    | 1                  | 1                 | 1                | 5     |
| Scalability          | 0             | 0             | -           | 1                    | 1                  | 1                 | 1                | 4     |
| Real-Time Visibility | 0             | 0             | 0           | -                    | 1                  | 1                 | 1                | 3     |
| Condition Response   | 0             | 0             | 0           | 0                    | -                  | 1                 | 1                | 2     |
| Energy Efficiency    | 0             | 0             | 0           | 0                    | 0                  | -                 | 1                | 1     |
| System Longevity     | 0             | 0             | 0           | 0                    | 0                  | 0                 | -                | 0     |

The team ranked objectives based on alignment with RFP-202601:

1. **Accessibility (7/28):** Measures the number of Universal Design principles satisfied and received the highest weight. The team prioritized this objective because RFP-202601 directly asks for a user-friendly solution; suitable designs should prioritize accessibility.
2. **Integrability (6/28):** Measures the number of systems being successfully controlled and ranked second. To satisfy the need for an integrated BMS at UVic, it is essential that the design displays multi-system capabilities.
3. **Scalability (5/28):** Estimates the per-zone sensor cost and came in fourth. The cost-effectiveness of scaling the design remains a critical factor for large-scale integration.
4. **Real-Time Visibility (4/28):** Measures the UI’s ability to reliably refresh, which is important for enabling live monitoring. However, this is not as essential as the design’s ability to provide a realistic and accessible solution to integrated building management.

5. **Condition Response (3/28):** Determines system response accuracy to programmed parameters, validating design function. This received a lower weight because system refinements could offer improvement.
6. **Energy Efficiency (2/28):** Assesses the design's potential reduction in energy use. While this aligns with UVic's current green goals, RFP-202601 seeks a solution to integrated control, which will ultimately aid in energy-optimization regardless of the design.
7. **System Longevity (1/28):** Reflects the number of components used in the MVP, attesting to anticipated maintenance complexity within a scaled version. This objective received lower consideration as the primary focus is on delivering an effective solution, not optimizing hardware reliability.